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# Towards provably secure asymmetric image encryption schemes

Naveed Ahmed Azam [a,b,*], Jianshen Zhu [a], Umar Hayat [b,*], Aleksandar Shurbevski [a]

[a] Department of Applied Mathematics and Physics, Kyoto University, Kyoto 606-8501, Japan
[b] Department of Mathematics, Quaid-i-Azam University, Islamabad 45320, Pakistan

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ABSTRACT

Image encryption schemes that use substitution box (S-box) generators, and dynamic S-boxes in particular, have attracted wide research interest recently. Typically, the security of S-box generators is tested empirically, and thus encryption schemes that use them do not have provable security against linear and differential attacks. With this work, we propose an asymmetric image encryption scheme with an S-box generator that has provable security against linear attacks, which is an important first step toward developing such schemes. Our generator is based on an integer linear programming (ILP) formulation that is feasible if and only if there exists an S-box with a given level of nonlinearity. We design a pseudo random number generator based on elliptic curves (ECs) which is used in our scheme to create confusion and diffusion in the cipher-images. Our scheme is highly sensitive to the keys and the plain-images, and hence it offers high security against differential attacks. Furthermore, our scheme allows users to use any size of random numbers and ECs, which makes it more flexible than existing schemes. Finally, we confirm through computational experiments that compared to existing state-of-the-art schemes, our scheme has superior cryptographic properties with low computational complexity, and hence is a good candidate for application in real-time systems.

## 1. Introduction

Owing to the intensive transmission of multimedia data over the internet, image encryption is a hot research topic in the field of data security. Images consist of highly correlated and redundant data, which makes it a challenging task to efficiently encrypt an image so that an adversary cannot get any useful information. Consequently, the literature is rich with various image encryption schemes (see, [5,9–12,14–18,21,22,24,25,27–30,33–37,41–43]).

Substitution-diffusion is a prominent framework used in image encryption schemes [1,3,4,8,12,13,18,20,28,31,33]. In this approach, dynamic substitution boxes (S-boxes) are used to create confusion in the plain-image, followed by some diffusion operations, which necessitate generating random numbers. Different mathematical structures are used to generate dynamic S-boxes and random numbers [3,4,12,13,18–20,28,32,33]. The security of the cipher-images obtained by these schemes is directly proportional to the cryptographic properties of the used S-boxes and random numbers.

Linear and differential attacks are the two most commonly used attacks applied on S-boxes. The resistance of an S-box to linear and differential attacks is quantified by its nonlinearity and differential approximation probability, respectively. Recently, Azam et al. [3], proposed some tests based on mathematically necessary conditions to quantify the dynamic behavior of an S-box generator. Several

* Corresponding author.
*E-mail addresses:* azam@amp.i.kyoto-u.ac.jp (N.A. Azam), umar.hayat@qau.edu.pk (U. Hayat).

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heuristics have been proposed to generate dynamic S-boxes with high nonlinearity by improving an initial S-box, see [2,38] and the references therein. However, these generators do not guarantee that they can output S-boxes with a desired nonlinearity, their dynamic behavior has not been tested nor have they been used in image encryption schemes.

Cryptography based on elliptic curves (ECs) provides high security with a comparatively small key size. Recently ECs have been used to reduce the size of the secret encryption keys in error correcting codes [40]. EC-based image encryption is also emerging as a new trend in the field of image security: Kumar et al. [22] designed an efficient image encryption scheme using DNA encoding and EC cryptography; Laiphrakpam and Khumanthem [23] combined ECs and chaotic systems to propose an efficient image cryptosystem; and Zhixiang et al. [42] used ECs and Toeplitz matrices to generate secure cipher-images from plain-images compressed by a wavelet transformation.

ECs are also used to develop substitution-diffusion image encryption schemes. Hayat and Azam [13] proposed the first substitution-diffusion image encryption scheme, where they generated dynamic S-boxes and random numbers based on ECs, and Azam et al. [4] proposed an improved EC-based image encryption scheme, whose computational complexity does not depend on the computation of points over the underlying EC. Aziz et al. [5] proposed an image encryption scheme by using ECs and the symmetric group. Haider et al. [12] designed ECs based S-boxes and used them for an image encryption application. Notably, Ibrahim and Alharbi [20] used a Henon map and dynamic S-boxes based on ECs to design an efficient image encryption algorithm.

However, existing encryption schemes suffer from several drawbacks:

(i) The cryptographic properties of S-box generators are not guaranteed:
A serious, yet common, drawback of all existing substitution-diffusion image encryption schemes is that their S-box generators do not guarantee that they can generate S-boxes with desired cryptographic properties such as nonlinearity and differential approximation probability, which are crucial to achieve high security against linear and differential attacks. In other words, the S-box generators that are used in image encryption do not have provable security, and thus the security of these encryption schemes is vulnerable to modern attacks.

(ii) Failure in the encryption process due to the S-box generator:
Although the scheme proposed by Hayat and Azam [13] can generate cipher-images with high security against modern attacks, it still (1) may fail to construct an S-box for a given set of input parameters, and thus the entire encryption process may fail; and (2) requires generating all points on an EC, which can be computationally costly when the underlying EC is large.

(iii) Small key size due to computation over EC:
The scheme due to Azam et al. [4] requires a precomputed EC, and the key size of this scheme is directly proportional to the size of the underlying prime. Therefore, because of computation and storage limitations, this scheme is not practical for large key sizes; and

(iv) Large size of random number sequences:
Furthermore, the aforementioned schemes use random numbers and ECs whose size is at least the size of the plain-image, which can further make them computationally costly with the increase in the size of the plain-image.

**Our Contributions** To address the aforementioned problems that exist with current state-of-the-art encryption schemes, we propose a novel five-phase public key image encryption scheme based on ECs. Our contributions can be summarized as follows:

- We propose an S-box generator that has provable security against linear attacks, as a first step in developing generators with provable security. Motivated from the fact that integer linear programs (ILPs) can be solved efficiently in practice, we design an ILP formulation that is feasible if and only if there exists a bijective S-box with a desired nonlinearity, as detailed in Section 3. Based on this generator, we design a confusion creation procedure for a plain-image, as detailed in Section 4, phase (III);
- Motivated by the highly random nature and security of ECs, we propose a new pseudo random number generator based on points that are computed recursively over ECs that are isomorphic to a given EC, as given in detail in Section 4, phases (I) and (II);
- We design a diffusion creation procedure that propagates a change in all pixel values of a given image in order to obtain a high level of security against differential attacks as detailed in Section 4, phase (IV);
- Motivated by the advantage of efficient point generation on isomorphic ECs, we define a mapping between pixels of a given image and the points of ECs that are isomorphic to a given EC, where the time complexity to compute points is constant. More precisely, this mapping is independent of the point computation over the ECs, as detailed in Section 4, phase (V);
- The scheme is flexible, in the sense that it can work with any size of random numbers and ECs, without any relation to the size of the plain-image, i.e., the random numbers and ECs used for a plain-image with $i$ pixels can also be used to obtain a secure cipher-image from a plain-image with $j$ pixels, $j > i$, as detailed in Section 4. Furthermore, the key space is user-defined, under the condition that keys are greater than $2^{100}$. That is, the user can arbitrarily increase the key space, which is not possible in some existing schemes [4,13], as detailed in Section 5.5.1;
- Most existing encryption schemes suffer from poor scalability with respect to the size of the plain-image. That is, the computational effort grows rapidly with the size of the plain-image. We design our scheme carefully so that it can create secure cipher-images of differently sized plain-images with a minimal difference in computational effort. Moreover, it can generate color cipher-images with a minimal difference in the computational complexity of generating grayscale cipher-images, as detailed in Section 5; and

- We conduct a rigorous analysis and comparison with a wide selection of existing image encryption schemes, [4,6,7,9,10,13,15,16,20–24,26,27,29,30,34,36] to show the superiority of the proposed scheme in terms of security and computational complexity, as detailed in Section 5.

The rest of the paper is organized as follows: In Section 2 we introduce some preliminaries. In Section 3 we present an ILP-based S-box generator and its detailed analysis. In Section 4 we propose a novel public key image encryption scheme. In Section 5 rigorous analysis, comparison and key features of the proposed scheme are discussed. Concluding remarks and future directions are given in Section 6.

## 2. Preliminaries

Throughout this draft, we use the symbol $\triangleq$ to denote/define some mathematical objects, e.g., the set of non-negative integers $\{z \in \mathbb{Z} \mid z \geq 0\}$ is denoted by $\mathbb{Z}_+$ and we write $\mathbb{Z}_+ \triangleq \{z \in \mathbb{Z} \mid z \geq 0\}$. We use the symbol $:=$ to assign a value to a variable, e.g., $x := 2$ means that the value of the variable $x$ is set to be 2; use the equality sign $=$ when two mathematical objects are identical, e.g., $4 = 5 - 1$; and use the $\equiv$ symbol when the modulo operation is applied, e.g., for two non-negative integers $h$ and $n$ such that $h = nq + r$ for some non-negative integers $q, r$ with $r < n$, we write either $h \equiv r \pmod{n}$ or $h \pmod{n} \equiv r$.

For two integers $a$ and $b$ with $a \leq b$, let $[a, b] \triangleq \{z \in \mathbb{Z} \mid a \leq z \leq b\}$. For an integer $z \in \mathbb{Z}_+$, let $\mathrm{bin}(z)$ denote the binary representation of $z$. For two vectors $\boldsymbol{v} = [v_1, v_2, \ldots, v_k]$ and $\boldsymbol{w} = [w_1, w_2, \ldots, w_k]$, let $\langle \boldsymbol{v}, \boldsymbol{w} \rangle$ denote the dot product of these vectors over the binary field, i.e., $\langle \boldsymbol{v}, \boldsymbol{w} \rangle \triangleq v_1 w_1 \oplus v_2 w_2 \oplus \cdots \oplus v_k w_k$, where $\oplus$ denotes the XOR operation. For a positive integer $n$, we denote by $\mathrm{GF}(2^n)$, the *Galois field extension* of the binary field $\mathrm{GF}(2)$. We denote by $(c)^{-1}$ the multiplicative inverse of an element $c$ in a field.

A bijective $(m \times m)$-S-box is a bijection $\sigma$ that maps $m$ input bits onto $m$ output bits, i.e., $\sigma : \mathrm{GF}(2^m) \to \mathrm{GF}(2^m)$, where the size of $\mathrm{GF}(2^m)$ is $2^m$.

**Elliptic curves over prime fields:** Let $p \geq 5$ be a prime, and $a, b \in [0, p-1]$ be two integers such that $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$. The *elliptic curve* $E_{p,a,b}$ over the prime field $\mathbb{F}_p$ is defined to be the set

$$E_{p,a,b} \triangleq \{(x, y) \in \mathbb{F}_p \times \mathbb{F}_p \mid y^2 \equiv x^3 + ax + b \pmod{p}\} \cup \{\infty\}.$$

We denote $E_{p,0,b}$ by $E_{p,b}$, for convenience. Henceforth, we call an elliptic curve over a prime field simply an elliptic curve (EC). Let $E_{p,a,b}$ and $E_{p,a',b'}$ be two ECs and $t \in [1, (p-1)/2]$ be an integer such that $at^4 \equiv a' \pmod{p}$ and $bt^6 \equiv b' \pmod{p}$. Then the mapping $\pi_t : E_{p,a,b} \to E_{p,a',b'}$ defined by $\pi_t(x, y) = (xt^2, yt^3) \pmod{p}$ is an isomorphism between the ECs, and we call $t$ an *isomorphism parameter* between the isomorphic ECs $E_{p,a,b}$ and $E_{p,a',b'}$.

**Group law over EC:** Let $G = (x_1, y_1)$ and $H = (x_2, y_2)$ be two points on an EC $E_{p,a,b}$. The *slope* $s(G, H)$ is defined to be

$$s(G,H) \triangleq \begin{cases} (y_1 - y_2)(x_1 - x_2)^{-1} & \text{if } G \neq H, x_1 \neq x_2 \\ (3x_1^2 + a)(2y_1)^{-1} & \text{if } G = H, y_1 \neq 0 \\ \infty & \text{if } \text{“}G \neq H, x_1 = x_2\text{”} \text{ or } \text{“}G = H, y_1 = 0\text{”} \end{cases} \pmod{p}. \tag{1}$$

The *group law* operation $+_{\mathrm{g}}$ on $G$ and $H$ results in a point $G +_{\mathrm{g}} H = (x_3, y_3)$ over $E_{p,a,b}$ defined as

$$(x_3, y_3) \triangleq \begin{cases} ((s(G,H))^2 - x_1 - x_2, -y_1 + s(G,H)(x_1 - x_3)) & \text{if } G \neq H, x_1 \neq x_2 \\ ((s(G,H))^2 - 2x_1, -y_1 + s(G,H)(x_1 - x_3)) & \text{if } G = H, y_1 \neq 0 \\ \infty & \text{if } \text{“}G \neq H, x_1 = x_2\text{”} \text{ or } \text{“}G = H, y_1 = 0\text{”} \end{cases} \pmod{p}. \tag{2}$$

Under the group law, $E_{p,a,b}$ is an abelian group with $\infty$ as the identity element. For an integer $n \leq \#(E_{p,a,b})$, we denote $\underbrace{G +_{\mathrm{g}} G +_{\mathrm{g}} \cdots +_{\mathrm{g}} G}_{n \text{ times}} \in E_{p,a,b}$ by $nG$.

**Nonlinearity:** For a cryptosystem to be secure against linear attacks, it is necessary to create a high level of confusion in the plain-text. For an $(m \times n)$-S-box $\sigma : \mathrm{GF}(2^m) \to \mathrm{GF}(2^n)$, the *nonlinearity* $\mathrm{NL}(\sigma)$ of $\sigma$ is defined to be

$$\mathrm{NL}(\sigma) \triangleq \min_{\substack{\alpha \in \mathrm{GF}(2^n) \setminus \{0\} \\ \beta \in \mathrm{GF}(2^m) \\ w \in \mathrm{GF}(2)}} \#(\{x \in \mathrm{GF}(2^m) \mid \langle \mathrm{bin}(\alpha), \mathrm{bin}(\sigma(x)) \rangle \neq \langle \mathrm{bin}(x), \mathrm{bin}(\beta) \rangle \oplus w\}). \tag{3}$$

Intuitively, the nonlinearity represents the Hamming distance of an S-box to any affine mapping on the Galois field $\mathrm{GF}(2^n)$, and is used as a measure of the S-box's capability to create confusion.

## 3. An ILP to generate S-boxes

In order to generate a bijective $(m \times m)$-S-box $\sigma$ that has nonlinearity at least $\ell$ for two integers $m \geq 4$ and $\ell \in [0, 2^{m-1} - 2^{m/2-1}]$, i.e., $\mathrm{NL}(\sigma) \geq \ell$, we propose an integer linear programming formulation $\mathrm{ILPNL}(m, \ell)$ that is feasible if and only if such an S-box exists. The intuition behind the variables and constraints of $\mathrm{ILPNL}(m, \ell)$ is explained below.

- There are $2^{2m}-2^m$ constants $y_{\alpha j} := \langle \text{bin}(\alpha), \text{bin}(j)\rangle$ in Eq. (4) to compute the linear Boolean functions $\langle \text{bin}(\alpha), \text{bin}(\sigma(x)\rangle$ that appear in Eq. (3), for $\alpha \in \text{GF}(2^m) \setminus \{0\}$ and $j \in \text{GF}(2^m)$.
- There are $2^{2(m+1)}$ constants $z_{\beta i w} := \langle \text{bin}(i), \text{bin}(\beta)\rangle \oplus w$ in Eq. (5) to compute the affine functions $\langle \text{bin}(x), \text{bin}(\beta)\rangle \oplus w$ that appear in Eq. (3), for $i \in \text{GF}(2^m)$, $\beta \in \text{GF}(2^m)$ and $w \in \text{GF}(2)$.
- There are $2^{2m}$ binary variables $x_{ij} \in \{0,1\}$ in Eq. (6), where $x_{ij} := 1$ (resp., $x_{ij} := 0$) implies that the input $i \in \text{GF}(2^m)$ is mapped (resp., not mapped) to $j \in \text{GF}(2^m)$.
- We introduce $2^{m+1}$ constraints in Eqs. (7) and (8) so that the resulting function is a bijection. Intuitively, Eq. (7) means that each input $i \in \text{GF}(2^m)$ is mapped to exactly one element $j \in \text{GF}(2^m)$ of the co-domain. By Eq. (8), each element of the co-domain $j \in \text{GF}(2^m)$ is mapped to exactly one input $i \in \text{GF}(2^m)$.
- There are $2 \cdot 2^m(2^m-1) = 2^{(m+1)}(2^m-1)$ nonlinearity constraints in Eq. (9), which ensure that the resulting S-box has nonlinearity at least $\ell$ if the ILPNL formulation is feasible. In Eq. (9), the expression $\sum_{i\in\text{GF}(2^m)} \sum_{j\in\text{GF}(2^m)} x_{ij}$ is used to encode the size of the set $\{x \in \text{GF}(2^m) \mid \langle \text{bin}(\alpha), \text{bin}(\sigma(x)\rangle \neq \langle \text{bin}(x), \text{bin}(\beta)\rangle \oplus w\}$ in Eq. (3), where the inequality $\langle \text{bin}(\alpha), \text{bin}(\sigma(x)\rangle \neq \langle \text{bin}(x), \text{bin}(\beta)\rangle \oplus w$ is encoded by $|y_{\alpha j} - z_{\beta i w}| \neq 0$. This means that for a fixed $i$ with $x_{ij} = 1$ and $|y_{\alpha j} - z_{\beta i w}| \neq 0$, the sum $\sum_{j\in\text{GF}(2^m)} x_{ij}|y_{\alpha j} - z_{\beta i w}|$ is effective and such $i$ will contribute to the nonlinearity. Note that Eq. (9) is linear in $x_{ij}$ since $|y_{\alpha j} - z_{\beta i w}|$ is a constant.

ILPNL$(m, \ell)$

Constants

$$y_{\alpha j} := \langle \text{bin}(\alpha), \text{bin}(j)\rangle, \qquad \forall \alpha \in \text{GF}(2^m) \setminus \{0\}, j \in \text{GF}(2^m), \tag{4}$$

$$z_{\beta i w} := \langle \text{bin}(i), \text{bin}(\beta)\rangle \oplus w, \qquad \forall i \in \text{GF}(2^m), \beta \in \text{GF}(2^m), w \in \text{GF}(2), \tag{5}$$

Variables

$$x_{ij} \in \{0,1\}, \qquad \forall i, j \in \text{GF}(2^m), \tag{6}$$

Constraints

$$\sum_{j\in\text{GF}(2^m)} x_{ij} = 1, \qquad \forall i \in \text{GF}(2^m), \tag{7}$$

$$\sum_{i\in\text{GF}(2^m)} x_{ij} = 1, \qquad \forall j \in \text{GF}(2^m), \tag{8}$$

$$\sum_{i,j\in\text{GF}(2^m)} x_{ij}|y_{\alpha j} - z_{\beta i w}| \geq \ell, \qquad \forall \alpha \in \text{GF}(2^m) \setminus \{0\}, \beta \in \text{GF}(2^m), w \in \text{GF}(2). \tag{9}$$

We give a necessary and sufficient condition for the feasibility of ILPNL in Lemma 1.

**Lemma 1.** *For two integers $m \geq 4$ and $\ell \in [0, 2^{m-1} - 2^{m/2-1}]$, the formulation* ILPNL$(m, \ell)$ *is feasible if and only if there exists an $(m \times m)$-S-box with nonlinearity at least $\ell$.*

**Proof.** If ILPNL$(m, \ell)$ is feasible, then this formulation outputs a mapping $\sigma : \text{GF}(2^m) \to \text{GF}(2^m)$ such that $\sigma(i) = j$ if and only if $x_{ij} = 1$. We show that $\sigma$ is a bijection and has $\text{NL}(\sigma) \geq \ell$. By Eq. (7), the mapping $\sigma$ is well-defined, i.e., for each input it has exactly one output. By Eq. (8), the mapping $\sigma$ is an injection. Thus, Eqs. (7) and (8), and the finite size of $\text{GF}(2^m)$ imply that $\sigma$ is a bijection. In Eq. (9), for fixed $\alpha, \beta$ and $w$, we compute the Hamming distance of Boolean functions and affine functions, and ensure that $\text{NL}(\sigma) \geq \ell$. Hence $\sigma$ is an $(m \times m)$-S-box with $\text{NL}(\sigma) \geq \ell$.

Conversely, let $\sigma$ be an $(m \times m)$-S-box with $\text{NL}(\sigma) \geq \ell$. We set $x_{ij} := 1$ (resp., $x_{ij} := 0$) if $\sigma(i) = j$ (resp., $\sigma(i) \neq j$). Then by the bijectivity of $\sigma$, the variables $x_{ij}$ satisfy Eqs. (7) and (8). Further, $\text{NL}(\sigma) \geq \ell$ implies that the variables $x_{ij}$ satisfy Eq. (9). Thus, the variables $x_{ij}$ defined by $\sigma$ give a feasible solution of the formulation ILPNL$(m, \ell)$, and hence ILPNL$(m, \ell)$ is feasible. □

By Lemma 1, our proposed ILP formulation can be used as an S-box generator to generate dynamic S-boxes with a desired nonlinearity value, if such an S-box exists.

An example of generating a $(4\times4)$-S-box with nonlinearity 4 using the ILP formulation ILPNL$(4, 4)$, giving all variables, constraints, and the output, is given in supplementary material S1_ILP_Sbox_example.

### 3.1. Results and discussion

By Lemma 1, for given integers $m \geq 4$ and $\ell \in [0, 2^{m-1} - 2^{m/2-1}]$, if the proposed ILP formulation ILPNL$(m, \ell)$ is feasible, then outputs a permutation with size $2^m$ and nonlinearity at least $\ell$. The infeasibility of ILPNL$(m, \ell)$ implies that there does not exist any such permutation.

We implemented the proposed ILP formulation in the C++ programming language using the `IBM ILOG CPLEX 12.8` solver. For experiments, we used a PC with Intel(R) Core(TM) i9-9900k CPU@2.60 GHz, 32 GB of memory, and Windows 10 Pro operating

**Table 1**
S-boxes generated by the proposed ILPNL formulations.

| $(m,\ell)$ | S-box | Time (sec.) |
|---|---|---|
| (4, 4) | [4, 0, 7, 2, 15, 10, 5, 12, 3, 1, 13, 9, 11, 6, 14, 8] | 0.028 |
| (5, 8) | [26, 19, 17, 6, 1, 27, 12, 31, 7, 18, 2, 25, 20, 14, 30, 16, 23, 21, 9, 13, 8, 5, 29, 22, 11, 24, 4, 3, 10, 0, 28, 15] | 1.885 |
| (5, 10) | [28, 2, 9, 5, 13, 12, 26, 22, 7, 24, 21, 6, 27, 10, 30, 3, 19, 29, 16, 8, 15, 25, 17, 1, 11, 23, 4, 0, 18, 31, 14, 20] | 5.18 |
| (6, 20) | [48, 38, 21, 25, 16, 0, 23, 36, 63, 30, 20, 31, 60, 50, 4, 49, 45, 34, 17, 41, 19, 27, 58, 56, 11, 33, 15, 24, 28, 13, 47, 12, 6, 51, 61, 46, 54, 22, 8, 40, 43, 35, 7, 9, 52, 62, 32, 42, 10, 37, 2, 59, 14, 39, 29, 26, 57, 18, 44, 1, 53, 5, 55, 3] | 30.411 |
| (6, 22) | [0, 32, 8, 17, 52, 49, 4, 11, 27, 41, 5, 19, 55, 48, 18, 46, 56, 37, 43, 21, 20, 7, 38, 35, 31, 15, 36, 45, 12, 39, 33, 47, 29, 42, 1, 63, 57, 53, 58, 59, 61, 16, 50, 13, 10, 3, 62, 24, 2, 23, 34, 26, 51, 9, 28, 40, 22, 25, 60, 6, 44, 14, 54, 30] | 273.686 |
| (7, 44) | [66, 22, 27, 28, 26, 21, 81, 12, 88, 19, 17, 46, 87, 97, 35, 58, 37, 40, 18, 68, 83, 42, 16, 74, 84, 80, 123, 104, 85, 60, 115, 79, 51, 6, 99, 25, 70, 10, 69, 3, 8, 15, 36, 55, 76, 120, 119, 111, 110, 126, 56, 102, 73, 86, 92, 5, 108, 1, 53, 9, 78, 39, 107, 2, 11, 13, 62, 113, 106, 67, 31, 117, 96, 72, 103, 89, 61, 33, 114, 90, 112, 75, 57, 118, 34, 65, 82, 77, 71, 122, 100, 14, 49, 47, 94, 105, 127, 52, 63, 59, 64, 41, 32, 98, 109, 43, 101, 116, 20, 38, 95, 124, 23, 121, 45, 54, 125, 93, 50, 91, 29, 4, 48, 44, 30, 0, 7, 24] | 555.159 |
| (7, 46) | [32, 20, 52, 31, 114, 121, 91, 29, 87, 1, 118, 77, 70, 43, 46, 93, 82, 67, 35, 124, 104, 15, 65, 12, 107, 116, 62, 37, 57, 115, 112, 48, 16, 92, 40, 83, 58, 127, 41, 123, 64, 17, 2, 45, 33, 96, 51, 6, 106, 27, 111, 68, 80, 7, 90, 103, 74, 97, 61, 108, 94, 79, 9, 59, 26, 95, 36, 30, 53, 105, 19, 75, 28, 47, 14, 99, 56, 11, 110, 122, 21, 102, 23, 81, 22, 55, 78, 63, 73, 109, 44, 125, 117, 84, 50, 4, 42, 86, 98, 100, 89, 5, 120, 10, 88, 0, 126, 71, 49, 76, 39, 25, 85, 69, 38, 34, 119, 8, 13, 24, 66, 72, 60, 101, 18, 3, 113, 54] | 1824.97 |
| (7, 48) | N.A. | T.O. |
| (8, 96) | N.A. | M.O. |

**Table 2**
Statistical results for the dynamic S-boxes generated by the proposed formulation.

| $(m,\ell)$ | (4, 4) | (5, 8) | (6, 18) |
|---|---|---|---|
| # generated S-boxes | 20,000 | 20,000 | 5,500 |
| # distinct S-boxes | 20,000 | 20,000 | 5,500 |
| Avg. difference | 15 | 31 | 63 |
| Avg. correlation | 0.209 | 0.144 | 0.101 |
| Avg. # fixed points | 1.005 | 1.010 | 1 |

system, where we set the time limit for the solver to 3600 seconds. For these experiments, we generated $(m \times m)$-S-boxes with nonlinearity at least $\ell$, $(m,\ell) \in \{(4,4),(5,8),(5,10),(6,20),(6,22),(7,44),(7,46),(7,48),(8,96)\}$, where $(4,4)$, $(5,10)$, and $(6,22)$ are the optimal pairs w.r.t. $m$. The output is given in Table 1. The resulting S-boxes are presented as vectors where the $i$-th entry denotes the output of an S-box for the integer $i$. For $(m,\ell) \in \{(4,4),(5,8),(5,10),(6,20),(6,22),(7,44),(7,46)\}$, the formulation was solved and the produced an S-box in less than 600 seconds. For $(m,\ell) = (7,48)$ (resp., $(m,\ell) = (8,96)$), the formulation was not solved due to a time out (resp., memory out). One of the main reasons for memory out is the exponential increase in the number of variables and the length of the expressions in Eqs. (7)–(9) with the increase in $m$. From Table 1, we observe that for a fixed $m$, the time to solve formulation ILPNL$(m,\ell)$ is directly proportional to $\ell$, and the ILP formulation is efficient for relatively small pairs $(m,\ell)$, such as $(4,4)$.

Azam et al. [3] introduced some tests to analyze if an S-box generator can generate highly dynamic S-boxes. In order to apply those tests against an S-box generator that uses the ILP formulation of Sec. 3, we introduce the following objective function in the proposed formulation

$$\max / \min \sum_{i,j \in \mathrm{GF}(2^m)} x_{ij} r_{ij}, \tag{10}$$

for random numbers $r_{ij} \in \mathbb{Z}$, $i,j \in \mathrm{GF}(2^m)$. We generated three sets of S-boxes with sizes 20,000, 20,000, and 5,500, by solving ILPNL$(m,\ell)$ with the objective function given in Eq. (10) for $(m,\ell) = (4,4)$, $(5,8)$, and $(6,18)$, respectively. For these sets of S-boxes we applied:

- variation analysis by computing the number of distinct S-boxes;
- sensitivity analysis by computing the average difference between S-boxes; and
- confusion analysis by computing the average correlation between S-boxes and average number of fixed points.

The results of these experiments are given in Table 2.

From Table 2, for each pair $(m,\ell) \in \{(4,4),(5,8),(6,18)\}$, we observe that all of the generated S-boxes were distinct, the average difference between the S-boxes was $2^m - 1$, the average correlation between the S-boxes did not exceed 0.209, and the average number of fixed points did not exceed 1.005. Hence, we can see that the proposed method can generate highly dynamic S-boxes with a given nonlinearity.

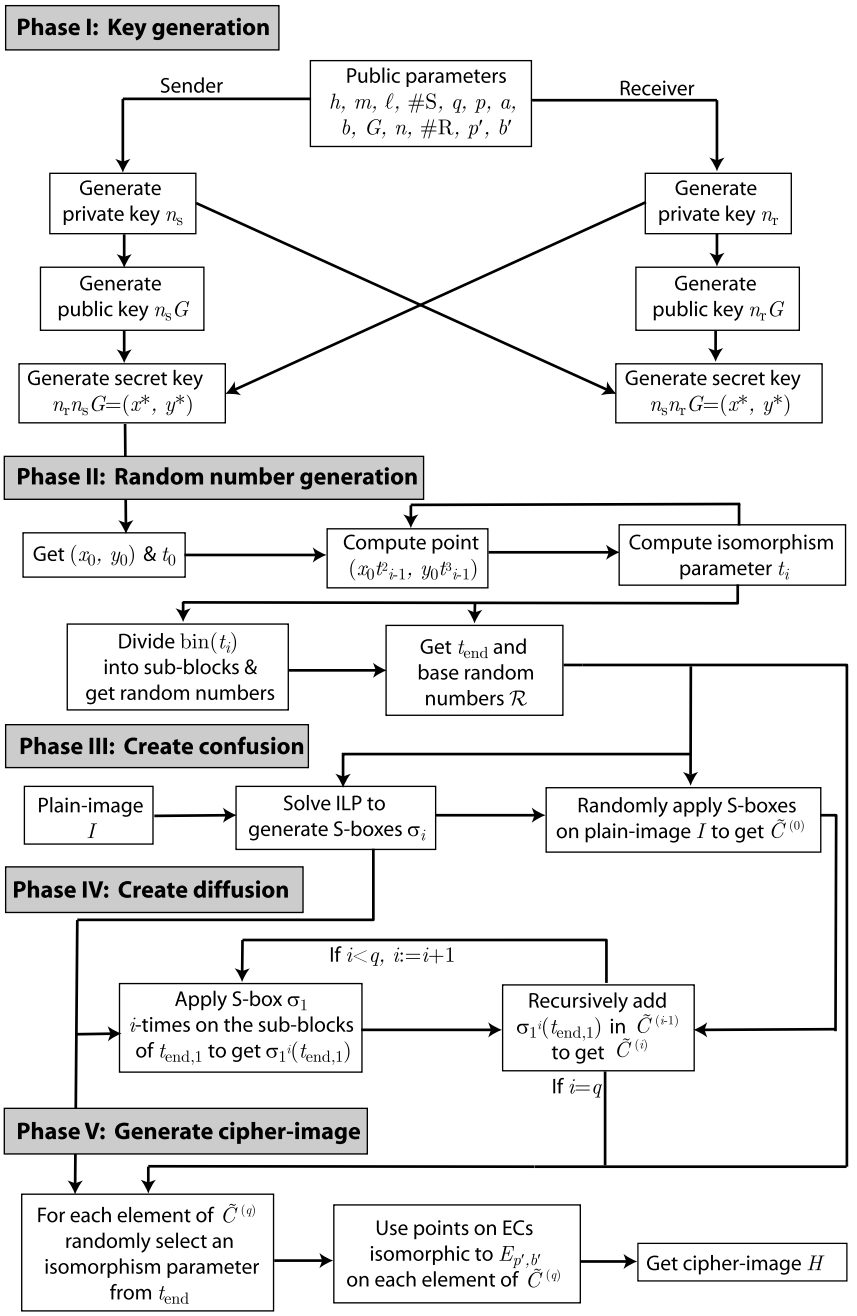



**Fig. 1.** Flowchart of the proposed scheme.

## 4. A public-key image encryption scheme

We discuss the details of our proposed encryption scheme in this section. There are five phases of our scheme, where

- the first phase (I) is to generate keys;
- the second phase (II) is to recursively generate some random numbers based on isomorphic ECs;
- the third phase (III) is to generate some dynamic S-boxes, which are then used to create confusion in the plain-image;
- the fourth phase (IV) is to create diffusion in the confused image by propagating a change in the pixel values; and
- the fifth phase (V) is to generate a cipher-image by using points over isomorphic ECs.

A flowchart of the proposed scheme is given in Fig. 1, and the proposed phases are explained in detail below.

**Phase I: Key Generation**

Our scheme consists of 13 public parameters on which the sender and receiver agree, two private keys known only to their respective owners, two public keys known to the sender and receiver, and a secret key. The generation process of these keys and their intuition is given below.

- Fix an integer $h$ that will be used in phase (II) to generate "base random numbers" by dividing a sequence of bits into sub-blocks;
- Fix an integer $\#\mathrm{R} \geq 16$ to determine the size of the sequence of base random numbers;
- Fix an integer $m \geq 4$, to determine the size of S-boxes to be used to create confusion in the plain-image in phase (III). Our proposed scheme can work for any $m$, however for computational efficiency, we use $m = 4$, which is commonly used in several cryptosystems;
- Fix an integer $\ell \in [0, 2^{m-1} - 2^{m/2-1}]$ to determine the nonlinearity of the S-boxes to be generated in phase (III);
- Fix an integer $\#\mathrm{S} \geq 2$ to determine the number of $(m \times m)$-S-boxes with nonlinearity at least $\ell$ that will be used in phase (III);
- Fix an integer $q \geq 2$ to determine the number of rounds of the diffusion operation in phase (IV);
- Generate a point on an EC that will be a secret key and will be used in phases (III)–(V). To generate such a secret key, we use the Diffie–Hellman protocol. We select an EC $E_{p,a,b}$ such that the size $|\mathrm{bin}(p)| \geq \max\{h, 100\}$, and a point $G \in E_{p,a,b}$ such that $nG = \infty$, where $n$ is the smallest positive integer with such property and $|\mathrm{bin}(n)| \geq 100$. The aim of the condition on the size of the prime is to select a prime large enough so that the Diffie-Hellman protocol can establish a secure key exchange protocol. The sender and receiver generate two numbers $n_{\mathrm{s}} < n$ and $n_{\mathrm{r}} < n$, and compute the points $n_{\mathrm{s}}G$ and $n_{\mathrm{r}}G$, respectively, which are then used to compute a point $(x^*, y^*) \triangleq n_{\mathrm{r}}n_{\mathrm{s}}G = n_{\mathrm{s}}n_{\mathrm{r}}G$;
- Fix an EC $E_{p',b'}$ such that $p' \equiv 2 \pmod 3$ and $p' \geq 17$. We use ECs that are isomorphic to $E_{p',b'}$ in phase (V) to create a cipher-image.

The sender and receiver agree on the public parameters $h, m, \ell, \#\mathrm{S}, q, p, a, b, G, n, \#\mathrm{R}, p'$, and $b'$. The integers $n_{\mathrm{s}}$ and $n_{\mathrm{r}}$ are the private keys of the sender and receiver, respectively; $n_{\mathrm{s}}G$ and $n_{\mathrm{r}}G$ are the public keys of the sender and receiver, respectively. The point $(x^*, y^*) \in E_{p,a,b}$ is a shared secret key of the sender and receiver.

Let $I$ be a $(u \times v)$-dimensional plain-image over the symbol set $[0, 2^{m^*} - 1]$ such that $uv \in [16, p']$.

**Phase II: Random Number Generation**

We propose a random number generator to generate base random numbers that we use in phase (III) to randomly generate S-boxes and to determine an assignment of S-boxes for each pixel of the plain-image; and in phase (V) to randomly generate isomorphism parameters and determine their assignment for each pixel. The idea of our generator is to start with the secret key $(x^*, y^*) \in E_{p,a,b}$ and compute an isomorphism parameter $t_0$. Then recursively compute isomorphism parameters $t_i$ from the points $(x_0 t_i^2, y_0 t_i^3)$ for some $(x_0, y_0) \in E_{p,a,b}$ which is obtained by Eq. (11). For each $t_i$, we divide $\mathrm{bin}(t_i)$ into sub-blocks using $h$ and get random numbers by converting each block into decimal form. The aim is to generate random numbers that are highly sensitive to the secret key $(x^*, y^*)$ so that a slight change in the secret key can result in an entirely different cipher-image to prevent differential attacks. We explain these steps below, where we denote by $P_{p,a,b}$, the smallest non-zero point on the EC $E_{p,a,b}$, i.e., for $P_{p,a,b} = (x, y)$ it holds that $(x, y) \neq (0, 0)$, and either "$x < x'$" or "$x = x'$ and $y < y'$" for all $(x', y') \in E_{p,a,b} \setminus \{(0, 0), (x, y)\}$. Note that the point $P_{p,a,b}$ is unique.

(II-i) We define the initial point $(x_0, y_0) \in E_{p,a,b}$ for our random number generator as follows

$$(x_0, y_0) \triangleq \begin{cases} P_{p,a,b} & \text{if } x^* = 0, y^* = 0 \\ (x^*, y^*) & \text{otherwise.} \end{cases} \tag{11}$$

We define the initial isomorphism parameter $t_0$ to determine the initial isomorphic EC as follows

$$t_0 \triangleq \begin{cases} x^* + y^* & \text{if } x^* + y^* \not\equiv 0 \\ x^* & \text{if } x^* + y^* \equiv 0, x^* \neq 0, y^* \neq 0 \\ x' + y' & \text{if } x^* = y^* = 0 \end{cases} \pmod p, \tag{12}$$

where $(x', y') = P_{p,a,b}$.

(II-ii) For an integer $i \geq 1$, we define the isomorphism parameter $t_i$ of the $i$-th isomorphic EC that will be used in the generation of the next random number as follows

$$t_i \triangleq \begin{cases} x_0 t_{i-1}^2 + y_0 t_{i-1}^3 & \text{if } x_0 t_{i-1}^2 + y_0 t_{i-1}^3 \not\equiv 0 \\ x_0 t_{i-1}^2 & \text{if } x_0 t_{i-1}^2 + y_0 t_{i-1}^3 \equiv 0, x_0 t_{i-1}^2 \not\equiv 0, y_0 t_{i-1}^3 \not\equiv 0 \\ (x_0 t_{i-1}^2)' + (y_0 t_{i-1}^3)' & \text{if } x_0 t_{i-1}^2 \equiv y_0 t_{i-1}^3 \equiv 0 \end{cases} \pmod p. \tag{13}$$

Observe that the points $(x_0 t_{i-1}^2, y_0 t_{i-1}^3)$ and $((x_0 t_{i-1}^2)', (y_0 t_{i-1}^3)') = P_{p,a,t_{i-1}^6 b}$ lie on the EC $E_{p,a,t_{i-1}^6 b}$ that is isomorphic to $E_{p,a,b}$.

(II-iii) For each $t_i$, $i \geq 1$, divide $\mathrm{bin}(t_i)$ into sub-blocks using $h$. More precisely, for the case $|\mathrm{bin}(t_i)| > h$, we divide $\mathrm{bin}(t_i)$ into $h - 1$ sub-blocks of size $\lceil |\mathrm{bin}(t_i)|/h \rceil$ and a sub-block of size $|\mathrm{bin}(t_i)| - \lceil |\mathrm{bin}(t_i)|/h \rceil (h-1)$. For the case $|\mathrm{bin}(t_i)| = h$ (resp., $|\mathrm{bin}(t_i)| < h$) divide $\mathrm{bin}(t_i)$ into $h$ (resp., $|\mathrm{bin}(t_i)|$) sub-blocks each of size 1. Finally, we get the base random numbers by converting each sub-block into its decimal form.

Let $\mathcal{R}$ denote the sequence obtained by listing the random numbers computed from $t_i$ with $i$ in ascending order, where $|\mathcal{R}| \geq \#\mathrm{R}$. We call $\mathcal{R}$ the sequence of base random numbers.

We remark that when $a = 0$, then $(0,0) \notin E_{p,a,b}$. This fact follows from $y^2 \equiv x^3 + ax + b \pmod{p}$ and $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$. Therefore for such a case we do not need to compute the point $P_{p,a,b}$ in Eqs. (11)–(13), which can be time consuming for large primes $p$.

**Phase III: Create Confusion**

We randomly generate #S $(m \times m)$-S-boxes, which are then used to create confusion in the plain-image. For random generation of S-boxes, we use the base random numbers in Eq. (10) and the ILPNL$(m, \ell)$ formulation. There are two main steps of this phase which are described below.

(III-i) Let $i \in [0, \#\mathrm{S} - 1]$ be an integer, and we want to generate $\sigma_i$, the $i$-th $(m \times m)$-S-box. To generate an $(m \times m)$-S-box using Eq. (10), we need $2^{2m}$ random numbers, and therefore to generate #S S-boxes we need $2^{2m} \cdot (\#\mathrm{S})$ random numbers. Let $\mathcal{R}'$ denote the sequence of random numbers obtained by concatenating the base random numbers $\mathcal{R}$ with itself. To generate the $i$-th S-box $\sigma_i$, we use $2^{2m}$ random numbers starting from the index $1 + 2^{2m} i$ from the sequence $\mathcal{R}'$ in Eq. (10) and the formulation ILPNL$(m, \ell)$.

(III-ii) Create confusion in the plain-image by using the S-boxes $\sigma_i$ where we use base random numbers $\mathcal{R}$ to determine the assignment of these S-boxes. Let $f$ be a linear indexing on the pixels of the plain-image $I$, i.e., $f : U \times V \to [1, uv]$ be a bijective function, where $U$ and $V$ denote the set of integers in $[1, u]$ and $[1, v]$, respectively. We consider the sequence $I_1 I_2 \dots I_{uv}$ of intensities of the plain-image, where $I_i$ denotes the intensity of the $i$-th pixel w.r.t. $f$ for $i \in [1, uv]$. We divide the sequence $\mathrm{bin}(I_1)\mathrm{bin}(I_2)\dots\mathrm{bin}(I_{uv})$ into sub-blocks of size $m$, and let $k$ denote the number of sub-blocks obtained. Convert each sub-block into decimal form and obtain the sequence $D(I) \triangleq d_1 d_2 \dots d_k$ over $[0, 2^m - 1]$, where $d_i$, $i \in [1, k]$ is the decimal form of the $i$-th sub-block. To determine the assignment of S-boxes for each entry of $D$, we define $\mathcal{R} \pmod{\#\mathrm{S}}$ to be the random sequence obtained by applying modulo #S on each entry of $\mathcal{R}$. Now scramble the values of $D(I)$ by applying $\sigma_j$ to get a scrambled binary sequence $C \triangleq c_1 c_2 \dots c_k$, such that for each $i \in [1, k]$, it holds that

$$c_i = \mathrm{bin}(\sigma_j(d_i)), \tag{14}$$

where $r_i \equiv j \pmod{\#\mathrm{R}}$ for the $i$-th entry $r_i$ of $\mathcal{R} \pmod{\#\mathrm{S}}$.

**Phase IV: Create Diffusion**

We create diffusion in the plain-image in $q$ rounds, where in each round we propagate a change in the output of the preceding round. For this purpose, in each round we generate a random number from an isomorphism parameter obtained in phase (II) and an S-box obtained in phase (III). We then recursively add the random number in the sequence of integers obtained in the previous round to create diffusion. This phase is executed in two steps, which are explained in detail below, after introducing some necessary notation.

Let $\sigma$ be an $(m \times m)$ S-box and $a \in [0, 2^m - 1]$. We denote by $\sigma^i(a)$ the application of $\sigma$ on $a$ $i$-times, i.e., $\sigma^i(a) \triangleq \sigma(\sigma^{i-1}(a))$, where $\sigma^0(a) \triangleq a$. Let $t_{\mathrm{end}}$ denote the last isomorphism parameter generated in phase (II), where we denote by $t_{\mathrm{end},i}$ the non-zero integer obtained from the $i$-th sub-block of $\mathrm{bin}(t_{\mathrm{end}})$ of size $\mathrm{bin}(p')$ and modulo $p'$. Let $t_{\mathrm{end},1,j}$ denote the integer obtained from the $j$-th sub-block of size $m$ of $t_{\mathrm{end},1}$. Let $\tilde{C}$ denote the sequence of integers over $[0, 2^{m^*} - 1]$ obtained by converting each sub-block of size $2^{m^*}$ of $C$ into decimal form. For an integer $i \leq q$, let $\tilde{C}^{(i-1)}$ denote the output of the $(i-1)$-th round of the diffusion phase, where we set $\tilde{C}^{(0)} \triangleq \tilde{C}$.

(IV-i) We generate a random number in the range $[0, p' - 1]$ by applying the S-box $\sigma_1$ on $t_{\mathrm{end},1}$ $i$-times to get another integer. More precisely, convert the sequence of integers over $[0, 2^m - 1]$ with $j$-th entry $\sigma_1(\sigma_1^{i-1}(t_{\mathrm{end},1,j}))$ into a binary sequence. Now apply modulo $p'$ on the decimal form of the resulting binary sequence to get a random integer $[0, p' - 1]$, where we denote this random integer by $\sigma_1^i(t_{\mathrm{end},1})$.

(IV-ii) Consider the sequence $\tilde{C}^{(i-1)}$ over $[0, 2^{m^*} - 1]$ obtained in the $(i-1)$-th round, where $\tilde{c}_j^{(i-1)}$ denotes its $j$-th entry. We get the output $\tilde{C}^{(i)}$ of the $i$-th round such that for the $j$-th entry $\tilde{c}_j^{(i)}$ of $\tilde{C}^{(i)}$, $j \geq 1$, we have:

$$\tilde{c}_j^{(i)} \triangleq \begin{cases} \tilde{c}_j^{(i-1)} + \sigma_1^i(t_{\mathrm{end},1}) & \text{if ``}j = 1 \text{ and } i \text{ is odd''} \text{ or ``}j = uv \text{ and } i \text{ is even''} \\ \tilde{c}_{j-1}^{(i)} + \tilde{c}_j^{(i-1)} & \text{if } j \geq 2 \text{ and } i \text{ is odd} \\ \tilde{c}_{j+1}^{(i)} + \tilde{c}_j^{(i-1)} & \text{if } j < uv \text{ and } i \text{ is even} \end{cases} \pmod{2^{m^*}}. \tag{15}$$

For the $i$-th round where $i$ is odd, the intuition of Eq. (15) is to get the first entry of $\tilde{C}^{(i)}$ by adding $\sigma_1^i(t_{\mathrm{end},1})$ in the first entry $\tilde{c}_1^{(i-1)}$ of $\tilde{C}^{(i-1)}$ under modulo $2^{m^*}$; and the $j$-th entry of $\tilde{C}^{(i)}$ is obtained from the addition of the $(j-1)$-th entry $\tilde{c}_{j-1}^{(i)}$ of $\tilde{C}^{(i)}$ and the $j$-th entry $\tilde{c}_j^{(i-1)}$ of $\tilde{C}^{(i-1)}$ under modulo $2^{m^*}$. For an even $i$, we perform the same procedure starting from the last entry of $\tilde{C}^{(i-1)}$. Observe that we are using the information of each entry of $\tilde{C}^{(i-1)}$ to get the output of the $i$-th round, and therefore the cipher-image generated by our scheme will have high resistance against differential attacks even if the secret key is not changed.

**Phase V: Generate a Cipher-Image**

We use points on some of ECs that are isomorphic to $E_{p',b'}$, the base random numbers and an S-box to generate a secure cipher-image. To determine isomorphic ECs, we select $2^m$ isomorphism parameters $t_{\mathrm{e},i}$, $i \in [0, 2^m - 1]$ as follows

$$t_{e,i} \triangleq \begin{cases} t_{\text{end},i+1} & \text{if } t_{\text{end},i+1} \text{ exists} \\ 1 & \text{otherwise.} \end{cases} \tag{16}$$

The idea is to diffuse each entry $\tilde{c}_i^{(q)}$ of $\tilde{C}^{(q)}$ by using a point on the EC $E_{p',t_{e,z}^6 b'}$ where the isomorphism parameter $t_{e,z}$ is determined by the application of S-box $\sigma_1$ on the base random numbers $\mathcal{R} \pmod{2^m}$. More precisely, we get a sequence $H$ over $[0, 2^{m^*}-1]$ whose $i$-th entry $h_i$ is defined as

$$h_i \triangleq (\tilde{c}_i^{(q)} + w + i') \pmod{2^{m^*}}, \tag{17}$$

where $i \equiv i' \pmod{p'}$, $(w, i') \in E_{p',t_{e,z}^6 b'}$, $z = \sigma_1(r_j \pmod{2^m})$, $\sigma_1$ is the $(m \times m)$-S-box obtained in phase (III), $r_j$ is the $j$-th entry of the random sequence $\mathcal{R}$ obtained in phase (II) and $i \equiv j \pmod{\#R}$.

Observe that due to the property of isomorphic ECs, it holds that

$$h_i = (t_{e,z}^2 w' \pmod{p'} + i' + \tilde{c}_i^{(q)}) \pmod{2^{m^*}}, \tag{18}$$

where $(w', (t_{e,z}^{-1})^3 i') \in E_{p',b'}$. Therefore by Eq. (18), we can compute the point $(w, i) \in E_{p',0,t_{e,z}^6 b'}$ in Eq. (17) in constant time, if the EC $E_{p',b'}$ and the multiplicative inverses $(t')^{-1}$ for each $t' \in [1, p'-1]$ are given. The right hand side of Eq. (18) is well-defined since for each $p' \equiv 2 \pmod 3$ and $i \in [0, p'-1]$, there exists a pair $(w, i') \in E_{p',b'}$ by the following theorem.

**Lemma 2** *([39]). For a prime $p \equiv 2 \pmod 3$, the MEC $E_{p,b}$ has exactly $p+1$ points and for each integer $y \in [0, p-1]$ there exists exactly one integer $x \in [0, p-1]$ such that $(x, y) \in E_{p,b}$.*

Finally, we get the cipher-image by converting $H$ into a matrix of dimension $u \times v$ by using the inverse of the indexing function $f$.

We give a step-by-step explanation of the entire encryption scheme applied on a plain-image of size $3 \times 3$ in the supplementary material S2_Encryption_example.

The public parameters $h, m, \ell, \#S, q, p, a, b, G, n, \#R, p'$, and $b'$, and the private key $n_r$ are known to the receiver. Therefore the receiver can uniquely decrypt the cipher-image by generating the secret key $(x^*, y^*)$, base random numbers $\mathcal{R}$, and S-boxes $\sigma_i$, and applying the phases (V), (IV), (III), (II), and (I).

## 5. Security analysis

The security of our scheme depends on the security of the secret key and the randomness in the cipher-image. The security of the secret key is ensured by the discrete logarithm problem over EC due to the Diffie–Hellman protocol. To show that the generated cipher-image is secure against several common attacks such as differential, statistical, and key attacks, we perform experiments to encrypt standard grayscale and color Lena images of size $512 \times 512$, All-white and All-black images of size $512 \times 512$, and standard grayscale and color images from the USC-SIPI Image Database[1] with size $i \times i$, $i \in \{256, 512, 1024\}$. For these experiments, we use a PC with Intel(R) Core(TM) i9-9900k CPU@2.60 GHz, 32 GB of memory, and Windows 10 Pro, and the following public parameters, public keys and secret keys:

- Number of sub-blocks of isomorphism parameters $h = 20$;
- Size of S-box $m = 4$;
- Minimum nonlinearity $\ell = 4$;
- Number of S-boxes $\#S = 5$;
- Size of the sequence of base random numbers $\#R = 400$;
- We use recommended EC $E_{p,a,b}$ secp521r1[2] for the generation of base random numbers, where $p = 2^{521} - 1$, $a =$ 01FFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFC, $b =$ 0051953EB9618E1C9A1F929A21A0B68540EEA2DA725B99B315F3B8B489918EF109E156193951EC7E937B 1652C0BD3BB1BF073573DF883D2C34F1EF451FD46B503F00;
- The base point $G = (x_G, y_G)$, where $x_G =$ 00C6858E06B70404E9CD9E3ECB662395B4429C648139053FB521F828AF606B4D3DB AA14B5E77EFE75928FE1DC127A2FFA8DE3348B3C1856A429BF97E7E31C2E5BD66, $y_G =$ 011839296A789A3BC0045C8A5FB42C7D1BD9 98F54449579B446817AFBD17273E662C97EE72995EF42640C550B9013FAD0761353C7086A272C24088BE94769FD16650;
- The order $n$ of $G$ is $n =$ 01FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFA51868783BF2F966B 7FCC0148F709A5D03BB5C9B8899C47AEBB6FB71E91386409;
- Private random numbers $n_s = 1$ and $n_r = 2$;
- Secret key $(x^*, y^*)$, where $x^* =$ 33C219024277E7E682FCB288148C282747403279B1CCC06352C6E5505D769BE97B3B204DA6EF55507 AA104A3A35C5AF41CF2FA364D60FD967F43E3933BA6D783D, $y^* =$ F4BB8CC7F86DB26700A7F3ECEEEED3F0B5C6B5107C4DA97740AB21 A29906C42DBBB3E377DE9F251F6B93937FA99A3248F4EAFCBE95EDC0F4F71BE356D661F41B02;

[1] SIPI, U. (2016). The USC-SIPI image database.

[2] Qu, M. (1999). Sec 2: Recommended elliptic curve domain parameters. *Certicom Res., Mississauga, ON, Canada, Tech. Rep. SEC2-Ver-0.6*.

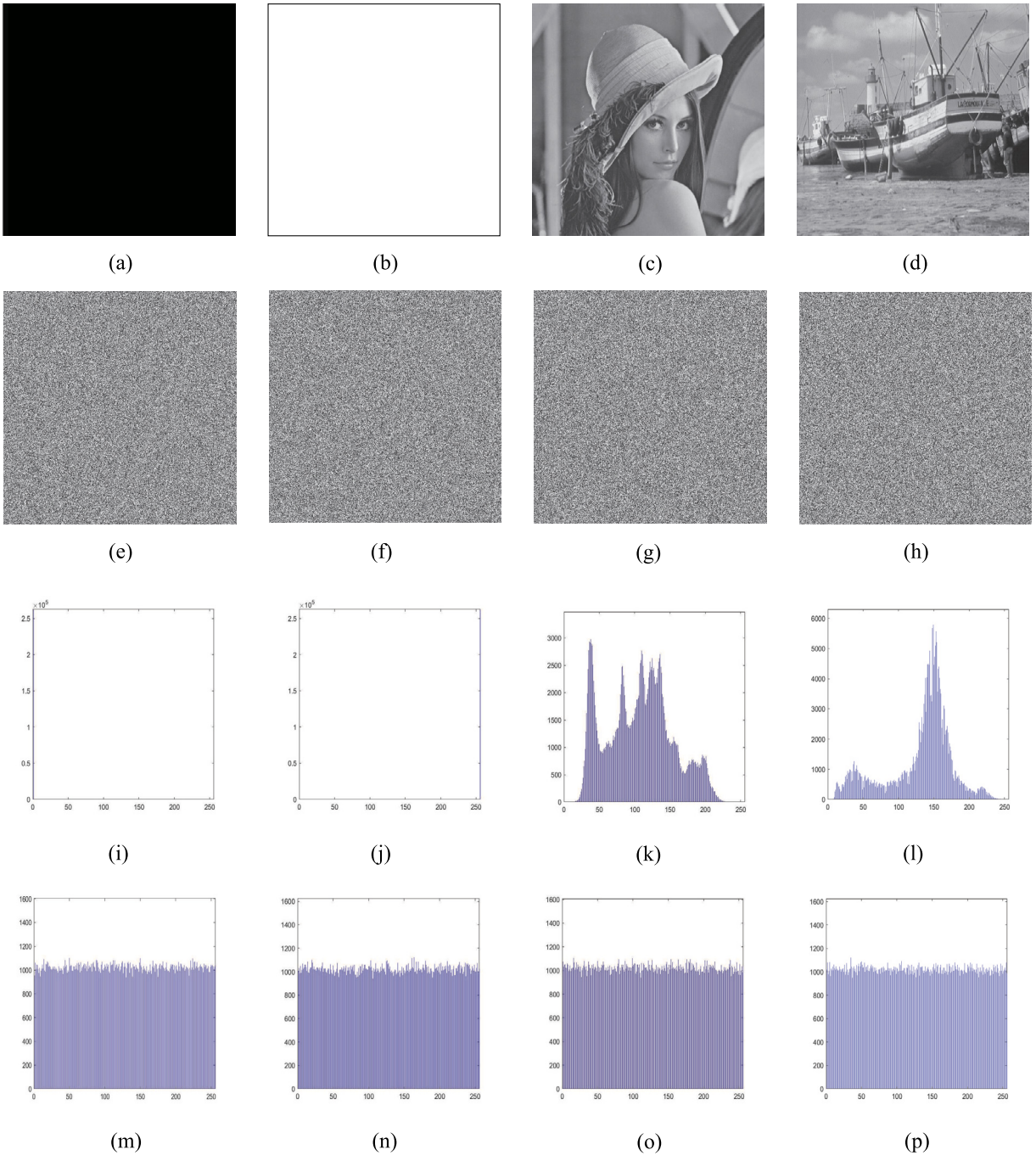



**Fig. 2.** Grayscale plain-images, cipher-images and their histograms: (a)–(d) Graysacle plain-images All-black, All-white, Lena and Boat with dimension $512 \times 512$, respectively; (e)–(h) Cipher-images of the plain-images in (a)–(d), respectively; (i)–(l) Histograms of the plain-images in (a)–(d), respectively; (m)–(p) Histograms of the cipher-images in (e)–(h), respectively.

– We use the public EC $E_{p',b'}$, where $p' = 1050737$ with $p' \equiv 2 \pmod 3$ and $b' = 1$. Since $E_{p',b'}$ is a public parameter therefore we pre-compute and store $E_{p',b'}$ and $(t')^{-1}$, for each $t' \in [1, (p'-1)/2]$.

We show graysacle plain-images All-black, All-white, Lena and Boat in Fig. 2(a)–(d), their cipher-images obtained by the proposed scheme in Fig. 2(e)–(h), histograms of the plain-images in Fig. 2(i)–(l), and histograms of the cipher-images in Fig. 2(m)–(p), respectively. Fig. 3 illustrates color plain-images Female, Lena, House, and San Francisco, their cipher-images obtained by the proposed scheme and their histograms.

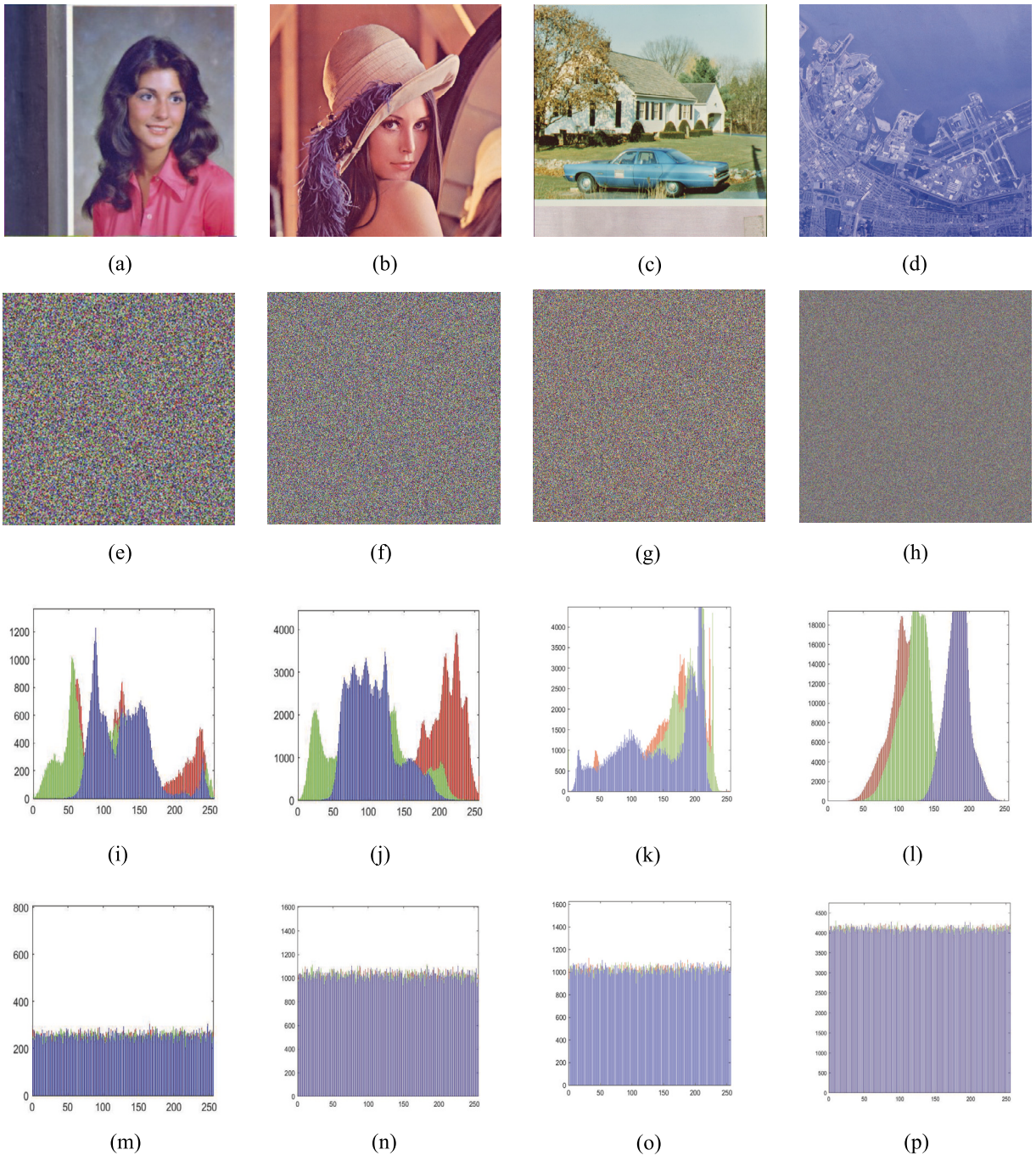



**Fig. 3.** Color plain-images, cipher-images and their histograms: (a)–(d) Color plain-images Female, Lena, House and San Francisco with dimension 256×256, 512×512, 512×512, and 1024×1024, respectively; (e)–(h) Cipher-images of the plain-images in (a)–(d), respectively; (i)–(l) Histograms of the plain-images in (a)–(d), respectively; (m)–(p) Histograms of the cipher-images in (e)–(h), respectively.

The results of our security analysis are discussed in the following sections.

### *5.1. Analysis of differential attacks*

Differential attacks are commonly used to decipher encrypted images. The aim of this attack is to analyze the change in the cipher-image when some pixels are changed in the plain-image. An encryption scheme is considered to be secure if it can generate a noticeably different cipher-image when only one pixel is changed in the plain-image. The resistance of a cryptosystem against

**Table 3**
Comparison of the passing ratio of NPCR and UACI of the proposed scheme with some of the existing schemes.

| Method | Ref. [4] | Ref. [9] | Ref. [13] | Ref. [14] | Ref. [15] | Ref. [24] | Ref. [25] |
|---|---|---|---|---|---|---|---|
| NPCR ratio | 0.96 | 0.96 | **0.83** | **0.92** | 0.96 | 0.96 | **0.2** |
| UACI ratio | 1.00 | N.A. | 1.00 | 1.00 | **0.96** | **0.96** | **0.04** |
| Method | Ref. [27] | Ref. [29] | Ref. [30] | Ref. [35] | Ref. [43] | Proposed | |
| NPCR ratio | 1.00 | 1.00 | **0.68** | **0.88** | **0.6** | 0.96 | |
| UACI ratio | 1.00 | 1.00 | **0.12** | **0.12** | N.A. | 1.00 | |

**Table 4**
Results of NPCR and UACI for differential attacks and key sensitivity for the cipher-images obtained by the proposed scheme for grayscale images.

| Grayscale image | Square dimension | Differential attacks | | Key sensitivity | |
|---|---|---|---|---|---|
| | | Avg. NPCR | Avg. UACI | NPCR | UACI |
| 5.1.09 | 256 | 99.6065 | 33.4500 | 99.6109 | 33.2835 |
| 5.1.10 | 256 | 99.6104 | 33.4163 | 99.6109 | 33.1994 |
| 5.1.11 | 256 | 99.5893 | 33.4170 | 99.6201 | 33.4249 |
| 5.1.12 | 256 | 99.6007 | 33.4850 | 99.6048 | 33.4983 |
| 5.1.13 | 256 | 99.6032 | 33.4706 | 99.6323 | 33.4920 |
| 5.1.14 | 256 | 99.5959 | 33.4654 | 99.5972 | 33.5687 |
| 5.2.08 | 512 | 99.6033 | 33.4653 | 99.6445 | 33.4896 |
| 5.2.09 | 512 | 99.5928 | 33.4660 | 99.6098 | 33.4672 |
| 5.2.10 | 512 | 99.6005 | 33.4907 | 99.6151 | 33.4058 |
| 5.3.01 | 1024 | 99.5969 | 33.4648 | 99.6136 | 33.4656 |
| 5.3.02 | 1024 | 99.5977 | 33.4612 | 99.6092 | 33.4501 |
| 7.1.01 | 512 | 99.6060 | 33.4674 | 99.6204 | 33.4450 |
| 7.1.02 | 512 | 99.5974 | 33.4833 | 99.6075 | 33.4654 |
| 7.1.03 | 512 | 99.6116 | 33.4838 | 99.6140 | 33.4716 |
| 7.1.04 | 512 | 99.6062 | 33.4534 | 99.6124 | 33.4545 |
| 7.1.05 | 512 | 99.5926 | 33.4621 | 99.6166 | 33.4946 |
| 7.1.06 | 512 | 99.5939 | 33.4802 | 99.6048 | 33.4743 |
| 7.1.07 | 512 | 99.5927 | 33.4268 | 99.6181 | 33.4185 |
| 7.1.08 | 512 | 99.5926 | 33.4793 | 99.6067 | 33.5156 |
| 7.1.09 | 512 | 99.5949 | 33.4781 | 99.6128 | 33.5601 |
| 7.1.10 | 512 | 99.5988 | 33.4785 | 99.6284 | 33.5074 |
| 7.2.01 | 1024 | 99.6065 | 33.4535 | 99.6094 | 33.4526 |
| Boat.512 | 512 | 99.6044 | 33.4553 | 99.6125 | 33.5207 |
| Gray21.512 | 512 | 99.6005 | 33.4723 | 99.6201 | 33.5078 |
| Ruler.512 | 512 | 99.5945 | 33.4578 | 99.6269 | 33.4608 |
| All-black | 512 | 99.6040 | 33.4911 | 99.6082 | 33.5217 |
| All-white | 512 | 99.5991 | 33.4598 | 99.6143 | 33.4870 |
| Lena | 512 | 99.6139 | 33.4631 | 99.6078 | 33.5292 |

differential attacks is measured by the pixel change rate (NPCR) and the unified average changing intensity (UACI). The mathematical expressions for NPCR and UACI are

$$\mathrm{NPCR} = \sum_{\substack{1\le r\le u,\\ 1\le s\le v}} \frac{W(r,s)}{P} \times 100, \tag{19}$$

$$\mathrm{UACI} = \sum_{\substack{1\le r\le u,\\ 1\le s\le v}} \frac{|E_I(r,s) - E_J(r,s)|}{P \cdot M} \times 100, \tag{20}$$

where $I$ and $J$ are two plain-images of the same size with a difference of one pixel over the symbol set $[0, M]$, $P$ is the number of pixels in $I$, $E_I$ and $E_J$ are the cipher-images of $I$ and $J$, respectively, and $W(r,s) = 0$ if $E_I(r,s) = E_J(r,s)$ and $W(r,s) = 1$ otherwise.

For an image of dimension $i \times i$, $i \in \{256, 512, 1024\}$, a cryptosystem has high security against differential attacks if it has NPCR (resp., UACI) at least 99.5693, 99.5893, and 99.5994 (resp., in the range [33.2824, 33.6447], [33.3730, 33.5541], and [33.4183, 33.5088]), respectively.

To perform this analysis, for each plain-image, we randomly selected a pixel position and randomly changed the pixel values. To get a better understanding of the behavior of the proposed scheme against differential attacks we randomly selected and changed pixel values 1000 times instead of only once. We calculated NPCR and UACI for 1000 trials where a single random pixel was changed in the image and then calculated the average of these values. These values are listed in Table 4 and Table 5 for grayscale and color plain-images, respectively.

We have compared the NPCR and UACI passing ratio of the proposed scheme with some of the existing schemes in Table 3. The NPCR passing ratio and UACI passing ratio of the proposed scheme is 96% and 100%, respectively. The NPCR passing ratio of the schemes [13,14,25,30,35,43] is less than 96%. Similarly, the UACI passing ratio of the schemes [15,24,25,30,35] is less than 100%.

**Table 5**
Results of NPCR and UACI for differential attacks and key sensitivity for the cipher-images obtained by the proposed scheme for color images.

| Color image | Channel | Square dimension | Differential attacks | | Key sensitivity | |
|---|---|---|---|---|---|---|
| | | | Avg. NPCR | Avg. UACI | NPCR | UACI |
| 4.1.01 | R | 256 | 99.5971 | 33.4227 | 99.6475 | 33.2484 |
| 4.1.01 | G | 256 | 99.6051 | 33.4863 | 99.6170 | 33.3778 |
| 4.1.01 | B | 256 | 99.5989 | 33.4423 | 99.5972 | 33.4686 |
| 4.1.02 | R | 256 | 99.5991 | 33.4766 | 99.6216 | 33.5113 |
| 4.1.02 | G | 256 | 99.5857 | 33.4063 | 99.6262 | 33.3440 |
| 4.1.02 | B | 256 | 99.6074 | 33.4305 | 99.5987 | 33.4212 |
| 4.1.03 | R | 256 | 99.6057 | 33.4398 | 99.6399 | 33.3557 |
| 4.1.03 | G | 256 | 99.5923 | 33.4610 | 99.6307 | 33.4405 |
| 4.1.03 | B | 256 | 99.5961 | 33.4958 | 99.6170 | 33.4644 |
| 4.1.04 | R | 256 | 99.6065 | 33.4709 | 99.6414 | 33.5665 |
| 4.1.04 | G | 256 | 99.6086 | 33.4320 | 99.6078 | 33.4809 |
| 4.1.04 | B | 256 | 99.6100 | 33.4429 | 99.5895 | 33.3995 |
| 4.1.05 | R | 256 | 99.5963 | 33.4884 | 99.6033 | 33.4194 |
| 4.1.05 | G | 256 | 99.6001 | 33.4197 | 99.6292 | 33.4341 |
| 4.1.05 | B | 256 | 99.6048 | 33.4380 | 99.6307 | 33.5029 |
| 4.1.07 | R | 256 | 99.6003 | 33.4337 | 99.6094 | 33.5322 |
| 4.1.07 | G | 256 | 99.5961 | 33.4464 | 99.6521 | 33.3519 |
| 4.1.07 | B | 256 | 99.6083 | 33.4422 | 99.6094 | 33.3285 |
| 4.2.01 | R | 512 | 99.5985 | 33.4543 | 99.6090 | 33.5024 |
| 4.2.01 | G | 512 | 99.5919 | 33.4825 | 99.6162 | 33.5654 |
| 4.2.01 | B | 512 | 99.6033 | 33.4875 | 99.6056 | 33.4888 |
| 4.2.05 | R | 512 | 99.6053 | 33.4650 | 99.5937 | 33.4058 |
| 4.2.05 | G | 512 | 99.6013 | 33.4327 | 99.6025 | 33.4490 |
| 4.2.05 | B | 512 | 99.6041 | 33.4703 | 99.6014 | 33.4707 |
| 4.2.06 | R | 512 | 99.6110 | 33.4713 | 99.6082 | 33.4921 |
| 4.2.06 | G | 512 | 99.5950 | 33.4753 | 99.5808 | 33.4621 |
| 4.2.06 | B | 512 | 99.5908 | 33.4680 | 99.6017 | 33.4716 |
| Lena | R | 512 | 99.6075 | 33.4687 | 99.6151 | 33.4811 |
| Lena | G | 512 | 99.6046 | 33.4607 | 99.6231 | 33.5138 |
| Lena | B | 512 | 99.6036 | 33.4773 | 99.6334 | 33.5921 |
| SanF | R | 1024 | 99.6032 | 33.4667 | 99.6163 | 33.4729 |
| SanF | G | 1024 | 99.6000 | 33.4616 | 99.6053 | 33.4860 |
| SanF | B | 1024 | 99.5998 | 33.4681 | 99.6050 | 33.4529 |

Thus it is evident from the comparison that the proposed scheme can offer better resistance against differential attacks compared to the schemes [13–15,24,25,30,35,43]. From Table 4 the range and average of NPCR (resp., UACI) of the cipher-images of the grayscale images is [99.5893, 99.6116] and 99.5997, respectively (resp., [33.4163, 33.4911] and 33.4643, respectively). Similarly from Table 5, the range and average of the NPCR (resp., UACI) of the cipher-images of the color images is [99.5857, 99.6110] and 99.6012, respectively (resp., [33.4063, 33.4958] and 33.4571, respectively). Almost all of the listed values are in the recommended ranges of NPCR and UACI, and therefore the proposed scheme can have a high level of security against differential attacks.

### 5.2. Analysis of information entropy

It is necessary for a cryptosystem to create a high level of randomness in the plain-image. Information entropy is used to measure the extent of randomness. The standard entropy $\mathrm{H}(X)$ for a given random variable $X$ over a set $\Omega$ is calculated as

$$\mathrm{H}(X) = -\sum_{\omega\in\Omega} \Pr(\omega)\log_2(\Pr(\omega)), \tag{21}$$

where $\Pr(\omega)$ is the probability of $\omega$ in $X$. From Eq. (21), it holds that $\mathrm{H}(X) \le \log(\#\Omega)$, e.g., $\mathrm{H}(X) \le 8$ if $\#\Omega = 256$. The randomness of a sequence of random numbers (resp., the confusion creation capability of a cryptosystem) is directly proportional to the entropy of the sequence (resp., the cipher-images generated by the cryptosystem).

In these experiments, we have generated a sequence of base random numbers $\mathcal{R}$ over $[0, 255]$ of size 400. The entropy of this sequence is 7.4578, which is in the acceptable range. For each plain-image and its corresponding cipher-image obtained by the proposed scheme, we have computed the information entropy and the results are listed in Tables 6 and 7. The range and average entropy of the grayscale images (resp., color images) is [0.0000, 7.5237] and 5.4106, respectively (resp., [5.2626, 7.6429] and

**Table 6**
Results of correlation and entropy analysis for graysacle images and their cipher-images obtained by the proposed scheme.

| Image | Correlation in plain-image | | | Correlation in cipher-image | | | Image entropy | |
|---|---|---|---|---|---|---|---|---|
| | Hori. | Diag. | Ver. | Hori. | Diag. | Ver. | Plain | Cipher |
| 5.1.09 | 0.9020 | 0.9038 | 0.9389 | -0.0028 | 0.0039 | -0.0018 | 6.7093 | 7.9973 |
| 5.1.10 | 0.9050 | 0.8260 | 0.8638 | 0.0025 | 0.0001 | 0.0064 | 7.3118 | 7.9971 |
| 5.1.11 | 0.9571 | 0.8927 | 0.9366 | 0.0039 | 0.0075 | -0.0003 | 6.4523 | 7.9975 |
| 5.1.12 | 0.9565 | 0.9389 | 0.9740 | -0.0029 | -0.0012 | -0.0002 | 6.7057 | 7.9976 |
| 5.1.13 | 0.8722 | 0.7563 | 0.8668 | -0.0024 | 0.0027 | -0.0024 | 1.5483 | 7.9971 |
| 5.1.14 | 0.9466 | 0.8541 | 0.8992 | -0.0043 | 0.0010 | 0.0025 | 7.3424 | 7.9970 |
| 5.2.08 | 0.9371 | 0.8562 | 0.8928 | -0.0032 | -0.0044 | -0.0024 | 7.2010 | 7.9993 |
| 5.2.09 | 0.9008 | 0.8035 | 0.8605 | -0.0013 | 0.0001 | 0.0007 | 6.9940 | 7.9993 |
| 5.2.10 | 0.9404 | 0.8981 | 0.9279 | -0.0011 | -0.0026 | -0.0001 | 5.7056 | 7.9993 |
| 5.3.01 | 0.9774 | 0.9672 | 0.9813 | 0.0010 | 0.0005 | 0.0013 | 7.5237 | 7.9998 |
| 5.3.02 | 0.9099 | 0.8591 | 0.9034 | 0.0003 | 0.0010 | 0.0002 | 6.8303 | 7.9998 |
| 7.1.01 | 0.9620 | 0.9075 | 0.9206 | 0.0008 | -0.0004 | 0.0010 | 6.0274 | 7.9994 |
| 7.1.02 | 0.9463 | 0.8963 | 0.9459 | 0.0016 | -0.0013 | -0.0032 | 4.0045 | 7.9993 |
| 7.1.03 | 0.9456 | 0.9017 | 0.9321 | 0.0024 | 0.0012 | -0.0008 | 5.4957 | 7.9992 |
| 7.1.04 | 0.9768 | 0.9560 | 0.9676 | -0.0005 | -0.0022 | 0.0005 | 6.1074 | 7.9993 |
| 7.1.05 | 0.9420 | 0.8934 | 0.9121 | -0.0017 | 0.0001 | -0.0002 | 6.5632 | 7.9992 |
| 7.1.06 | 0.9402 | 0.8861 | 0.9062 | 0.0053 | 0.0025 | -0.0021 | 6.6953 | 7.9992 |
| 7.1.07 | 0.8862 | 0.8392 | 0.8778 | -0.0007 | 0.0003 | 0.0050 | 5.9916 | 7.9993 |
| 7.1.08 | 0.9577 | 0.9220 | 0.9293 | 0.0008 | 0.0011 | 0.0028 | 5.0534 | 7.9992 |
| 7.1.09 | 0.9657 | 0.9168 | 0.9305 | -0.0009 | 0.0003 | -0.0009 | 6.1898 | 7.9994 |
| 7.1.10 | 0.9643 | 0.9313 | 0.9474 | 0.0013 | 0.0031 | -0.0008 | 5.9088 | 7.9992 |
| 7.2.01 | 0.9647 | 0.9450 | 0.9470 | 0.0002 | -0.0010 | 0.0007 | 5.6415 | 7.9998 |
| Boat.512 | 0.9381 | 0.9222 | 0.9713 | -0.0012 | 0.0005 | 0.0008 | 7.1914 | 7.9993 |
| Gray21.512 | 0.9965 | 0.9964 | 0.9998 | -0.0025 | -0.0001 | 0.0000 | 4.3923 | 7.9994 |
| Ruler.512 | 0.4542 | -0.0290 | 0.4643 | -0.0012 | -0.0024 | -0.0019 | 0.5000 | 7.9992 |
| All-black | – | – | – | 0.00035 | -0.0011 | -0.00165 | 0.0000 | 7.9993 |
| All-white | – | – | – | 0.0001 | 0.0005 | 0.0004 | 0.0000 | 7.9994 |
| Lena | 0.9761 | 0.9639 | 0.9877 | -0.0016 | 0.0052 | -0.0006 | 7.3944 | 7.9993 |

6.5252, respectively). From Table 6 (resp., Table 7) the entropy of cipher-images corresponding to the grayscale (resp., color) images is in the interval [7.9970, 7.9998] (resp., [7.9965, 7.9998]) with average entropy 7.9989 (resp., 7.9982). Observe that the entropy of the cipher-images is significantly improved when compared to their plain-images and is very close to the optimal values. Hence the proposed encryption scheme can create a highly random cipher-image.

We compared the entropy of the cipher-images for the grayscale and color Lena image with dimension $512 \times 512$ obtained by the proposed schemes and some existing schemes in Table 8. We observed that the cipher-image generated by the proposed scheme has entropy greater than that of the cipher-images obtained by [4,23,25,30,34,35]. Therefore the proposed scheme can generate a highly confused cipher-image as compared to the existing schemes.

### 5.3. Correlation analysis

In an image, adjacent pixels are highly correlated and it is crucial for a cryptosystem to break the correlation so that a cryptanalyst cannot draw any useful information from the cipher-image. For two random variables $X$ and $Y$ of size $P$, we compute the correlation coefficient $\mathrm{CC}(X,Y)$ in the horizontal, diagonal and vertical directions of adjacent pixels using

$$\mathrm{CC}(X,Y) = \frac{\frac{1}{P}\sum_{i=1}^{P}(x_i - \overline{X})(y_i - \overline{Y})}{\sqrt{K(X)K(Y)}}, \tag{22}$$

where $x_i$ and $y_i$ are the $i$-th element of the sequences $X$ and $Y$, respectively, $K(X) = \frac{1}{P}\sum_{i=1}^{P}(x_i - \overline{X})^2$, $K(Y) = \frac{1}{P}\sum_{i=1}^{P}(y_i - \overline{Y})^2$, and $\overline{X}$ and $\overline{Y}$ are the mean values of $X$ and $Y$, respectively.

For each grayscale image (resp., color image) and its corresponding cipher-image, we computed the correlation coefficients of pixels in the horizontal, diagonal and vertical directions and the results are listed in Table 6 (resp., Table 7).

For grayscale images, the correlation range (resp., average) in the horizontal, diagonal and vertical direction is [0.4542, 0.9965] (resp., 0.9239), [0.0290, 0.9964] (resp., 0.8639) and [0.4643, 0.9998] (resp., 0.9110), respectively, where we computed the average over the absolute values. For the cipher-images corresponding to the grayscale images, the correlation range (resp., average) in horizontal, diagonal and vertical direction is [0.0001, 0.0053] (resp., 0.0017), [0.0001, 0.0075] (resp., 0.0017), and [0.0000, 0.0064] (resp., 0.0015), respectively. Clearly, the cipher-images have very low correlation as compared to their corresponding grayscale images. Similarly, the color images have correlation range (resp., average) in horizontal, diagonal and vertical direction in the interval [0.7444, 0.9936] (resp., 0.9568), [0.6314, 0.9894] (resp., 0.9253) and [0.7442, 0.9951] (resp., 0.9524), respectively. Furthermore, the cipher-images of the color images have range (resp., average) of correlation in horizontal, diagonal and vertical direction in the

**Table 7**
Results of correlation and entropy analysis for color images and their cipher-images obtained by the proposed scheme.

| Color image | Channel | Correlation in plain-image | | | Correlation in cipher-image | | | Image entropy | |
|---|---|---|---|---|---|---|---|---|---|
| | | Hori. | Diag. | Ver. | Hori. | Diag. | Ver. | Plain | Cipher |
| 4.1.01 | R | 0.9729 | 0.9488 | 0.9626 | -0.0070 | 0.0005 | 0.0032 | 6.4200 | 7.9974 |
| 4.1.01 | G | 0.9719 | 0.9499 | 0.9646 | 0.0002 | -0.0031 | -0.0037 | 6.4457 | 7.9972 |
| 4.1.01 | B | 0.9584 | 0.9376 | 0.9517 | -0.0071 | -0.0057 | 0.0023 | 6.3807 | 7.9977 |
| 4.1.02 | R | 0.9493 | 0.9178 | 0.9562 | 0.0058 | 0.0085 | -0.0019 | 6.2499 | 7.9972 |
| 4.1.02 | G | 0.9308 | 0.9003 | 0.9534 | -0.0040 | -0.0082 | -0.0016 | 5.9642 | 7.9969 |
| 4.1.02 | B | 0.9178 | 0.8890 | 0.9442 | 0.0007 | 0.0000 | -0.0069 | 5.9309 | 7.9974 |
| 4.1.03 | R | 0.9779 | 0.9166 | 0.9323 | -0.0072 | 0.0042 | 0.0007 | 5.7150 | 7.9965 |
| 4.1.03 | G | 0.9748 | 0.8991 | 0.9148 | -0.0008 | 0.0047 | 0.0046 | 5.3738 | 7.9973 |
| 4.1.03 | B | 0.9726 | 0.9004 | 0.9168 | -0.0002 | -0.0126 | -0.0056 | 5.7117 | 7.9973 |
| 4.1.04 | R | 0.9786 | 0.9683 | 0.9878 | -0.0053 | -0.0085 | 0.0023 | 7.2549 | 7.9973 |
| 4.1.04 | G | 0.9660 | 0.9509 | 0.9820 | -0.0021 | 0.0000 | 0.0060 | 7.2704 | 7.9974 |
| 4.1.04 | B | 0.9523 | 0.9309 | 0.9717 | -0.0024 | -0.0057 | -0.0014 | 6.7825 | 7.9971 |
| 4.1.05 | R | 0.9671 | 0.9129 | 0.9354 | 0.0004 | 0.0058 | -0.0037 | 6.4311 | 7.9971 |
| 4.1.05 | G | 0.9805 | 0.9322 | 0.9475 | -0.0011 | 0.0016 | 0.0001 | 6.5389 | 7.9972 |
| 4.1.05 | B | 0.9820 | 0.9626 | 0.9750 | -0.0073 | -0.0015 | -0.0030 | 6.2320 | 7.9968 |
| 4.1.07 | R | 0.9745 | 0.9538 | 0.9763 | -0.0063 | -0.0049 | 0.0030 | 5.2626 | 7.9969 |
| 4.1.07 | G | 0.9757 | 0.9603 | 0.9801 | 0.0052 | -0.0006 | -0.0058 | 5.6947 | 7.9972 |
| 4.1.07 | B | 0.9890 | 0.9799 | 0.9880 | -0.0006 | 0.0049 | 0.0037 | 6.5464 | 7.9971 |
| 4.2.01 | R | 0.9936 | 0.9894 | 0.9951 | 0.0014 | 0.0027 | -0.0010 | 6.9481 | 7.9994 |
| 4.2.01 | G | 0.9812 | 0.9712 | 0.9872 | -0.0019 | -0.0021 | -0.0005 | 6.8845 | 7.9993 |
| 4.2.01 | B | 0.9826 | 0.9652 | 0.9791 | 0.0056 | -0.0068 | -0.0014 | 6.1265 | 7.9994 |
| 4.2.05 | R | 0.9726 | 0.9354 | 0.9575 | -0.0008 | 0.0023 | 0.0012 | 6.7178 | 7.9994 |
| 4.2.05 | G | 0.9578 | 0.9327 | 0.9678 | -0.0021 | 0.0000 | 0.0034 | 6.7990 | 7.9993 |
| 4.2.05 | B | 0.9640 | 0.9176 | 0.9376 | 0.0012 | -0.0004 | -0.0003 | 6.2138 | 7.9992 |
| 4.2.06 | R | 0.9558 | 0.9419 | 0.9540 | -0.0004 | 0.0005 | -0.0005 | 7.3124 | 7.9992 |
| 4.2.06 | G | 0.9715 | 0.9530 | 0.9662 | -0.0005 | 0.0017 | 0.0029 | 7.6429 | 7.9993 |
| 4.2.06 | B | 0.9710 | 0.9529 | 0.9693 | 0.0003 | -0.0005 | 0.0024 | 7.2136 | 7.9992 |
| Lena | R | 0.9753 | 0.9636 | 0.9869 | -0.0009 | -0.0037 | -0.0044 | 7.2622 | 7.9993 |
| Lena | G | 0.9745 | 0.9627 | 0.9869 | -0.0015 | 0.0025 | -0.0002 | 7.5898 | 7.9992 |
| Lena | B | 0.9530 | 0.9334 | 0.9736 | -0.000795 | 0.0008 | -0.0006 | 6.9843 | 7.9995 |
| SanF | R | 0.9058 | 0.8558 | 0.9052 | 0.0002 | 0.0001 | -0.0025 | 6.8181 | 7.9998 |
| SanF | G | 0.8785 | 0.8173 | 0.8785 | -0.0004 | -0.0012 | 0.0011 | 6.6249 | 7.9998 |
| SanF | B | 0.7444 | 0.6314 | 0.7442 | -0.0009 | -0.0002 | 0.0009 | 5.9899 | 7.9998 |

interval [0.0002, 0.0073] (resp., 0.0025), [0.0000, 0.0126] (resp., 0.0023), and [0.0001, 0.0069] (resp., 0.0025), respectively. These experimental results suggest that the proposed scheme can generate highly secure cipher-images.

We compared the correlation of the cipher-images for the image Lena obtained by the proposed scheme and other schemes in Table 8. The proposed scheme generated a cipher-image which has considerably lower correlation coefficient compared to that of the cipher-images generated by the scheme [14,20,22–25,29,30,35,37]. This further justifies our claim that the proposed scheme can generate highly confused cipher-images compared to the existing schemes.

### 5.4. Histogram analysis

For an encryption scheme, it is necessary to create a cipher image where all pixel values are used uniformly. We have computed the histograms of all plain-images and their corresponding cipher-images obtained by the proposed scheme. The histograms of the grayscale plain-images All-black, All-white, Lena and Boat (resp., color plain-images Female, Lena, House and San Francisco) are given in Fig. 2(i)–(l) (resp., Fig. 3(i)–(l)) which are highly non-uniform. However the histograms of the cipher-images of these plain-images are given in Figs. 2(m) (p) and 3(m) (p) which are almost uniform. Thus, the proposed scheme has high security against statistical attacks.

### 5.5. Analysis of the secret key

#### 5.5.1. Key space

Brute-force attacks are commonly used to decrypt the cipher-image where a cryptanalyst tries to enumerate all possible secret keys. Thus, an encryption scheme must have a large key space to have a high level of resistance against brute-force attacks. An encryption scheme can resist brute-force attacks if its key space is at least $2^{100}$.

**Table 8**
Comparison of the correlation and entropy of the cipher-images for grayscale and color image Lena with dimension $512 \times 512$ obtained by the proposed scheme and other existing schemes.

| Image | Method | Correlation of Cipher-image | | | Entropy |
|---|---|---|---|---|---|
| | | Horizontal | Diagonal | Vertical | |
| Grayscale | Ref. [4] | -0.0003 | -0.0005 | 0.0005 | **7.9991** |
| | Ref. [7] | **0.0025** | 0.0029 | **0.0027** | 7.9993 |
| | Ref. [13] | 0.0009 | **0.0097** | **-0.0013** | 7.9993 |
| | Ref. [14] | **-0.0353** | **0.0286** | **-0.0249** | 7.9993 |
| | Ref. [20] | **0.0021** | **0.1170** | **0.0125** | 7.9994 |
| | Ref. [21] | -0.0013 | **0.0080** | **-0.0094** | 7.9993 |
| | Ref. [24] | **0.0287** | **-0.0322** | **-0.0113** | 7.9993 |
| | Ref. [25] | **-0.0299** | **0.0076** | **0.0408** | **7.9634** |
| | Ref. [26] | **0.0019** | 0.0024 | **0.0010** | 7.9993 |
| | Ref. [27] | **-0.0021** | -0.0051 | **-0.0040** | NA |
| | Ref. [29] | **0.0039** | **0.0059** | **-0.0050** | 7.9994 |
| | Ref. [30] | **0.0335** | **-0.0174** | **-0.0295** | **7.9976** |
| | Ref. [34] | -0.0005 | 0.0012 | **0.0007** | **7.9972** |
| | Ref. [35] | **-0.0059** | **-0.0146** | **0.0211** | **7.9992** |
| | Ref. [36] | 0.0016 | 0.0020 | **0.0014** | 7.9993 |
| | Proposed | -0.0016 | 0.0052 | -0.0006 | 7.9993 |
| Image | Method | Correlation of Cipher-image | | | Entropy |
| | | Horizontal | Diagonal | Vertical | |
| Color | Ref. [22] | **0.0035** | **-0.0410** | -0.0017 | N.A. |
| | | **-0.0097** | **-0.0085** | **0.0106** | N.A. |
| | | **0.0185** | **-0.0174** | **0.0011** | N.A. |
| | Ref. [23] | **-0.0031** | **0.0055** | **-0.0060** | **7.9992** |
| | | **-0.0067** | **-0.0048** | **0.0127** | 7.9992 |
| | | -0.0005 | **-0.0016** | **-0.0041** | **7.9992** |
| | Ref. [37] | **-0.0127** | **0.0067** | **0.0060** | N.A. |
| | | **-0.0075** | **-0.0068** | **-0.0078** | N.A. |
| | | -0.0007 | **0.0042** | **0.0026** | N.A. |
| | Proposed | -0.0009 | -0.0037 | -0.0044 | 7.9993 |
| | | -0.0015 | 0.0025 | -0.0002 | 7.9992 |
| | | -0.0008 | 0.0008 | -0.0006 | 7.9995 |

There are three main brute-force attacks that a cryptanalyst can try to apply on our scheme, which are given below. For each of these methods, we show that the cryptanalyst needs to try more than $2^{100}$ possibilities to decipher the cipher-image.

(1) The cryptanalyst tries all points $(x, y)$ on the EC $E_{p,a,b}$ that can be generated by the base point $G = (x_G, y_G)$. There are $n$ such points that the cryptanalyst needs to try. However, we selected an EC and a base point $G$ such that $|\text{bin}(n)| \geq 100$ as a requirement of the input parameters. For example, for the order $n$ of the base point $G$ that is used in our experiments, it holds that $|\text{bin}(n)| = 521$. Thus the proposed scheme is secure against this kind of brute-force attacks.

(2) Recall that for the secret key $(x^*, y^*)$ it holds that $(x^*, y^*) = n_r n_s G = n_s n_r G$, where $n_s G$ and $n_r G$ are public keys of the sender and the receiver, respectively. Thus the cryptanalyst can compute the secret key by computing either $n_s$ or $n_r$. In other words, the cryptanalyst needs to solve the discrete logarithm problem which is proven to be computationally difficult with the currently available computational technologies. Hence, the cryptanalyst cannot get the secret key with this method in a reasonable time.

(3) The cryptanalyst may exhaustively try all possible random numbers, S-boxes, and isomorphism parameters without using a secret key. For this purpose, the cryptanalyst needs to try:
   – all possible arrangements of length #S of $(m \times m)$-S-boxes with nonlinearity at least $\ell$ in phase (III). Let $\alpha$ be the number of all possible $(m \times m)$-S-boxes with nonlinearity at least $\ell$. Then the number of all possible arrangements of length #S is $(\#S)! \cdot \binom{\alpha}{\#S}$.
   – all possible random sequences $\mathcal{R} \pmod{\#S}$ to determine the assignment of S-boxes in Eq. (14), where for the size #R of each sequence $\mathcal{R}$ it holds that $\#R \leq uv$ and $m \leq m^*$. There are $(\#S)^{(\#R)}$ such sequences.
   – in each round of phase (IV), the cryptanalyst needs to try all possible positive integers that are at most $p' - 1$. There are $q$ rounds in phase (IV), and therefore there are $(p' - 1)^q$ choices of integers in phase (IV).
   – to reverse the decryption, the cryptanalyst needs a sequence of integers over $[1, p' - 1]$ with size $2^m$ and a sequence $\mathcal{R} \pmod{2^m}$ of random numbers of length #R. Thus there are $(p' - 1)^\beta \cdot \beta^{(\#R)}$ choices, where $\beta = 2^m$ for such sequences.

   This implies that a cryptanalyst needs to try $\gamma$ choices

$$\gamma \triangleq (\#S)! \cdot \binom{\alpha}{\#S} \cdot (\#S)^{(\#R)} \cdot (p' - 1)^{q+\beta} \cdot \beta^{(\#R)}. \tag{23}$$

   We observe that for the minimum value of each parameter $m = 4, \ell = 4$, $\alpha = 20,000$, $\beta = 16$, $q = 2$, $p' = 17$ and $\#R = 16$, it holds that $\gamma > 2^{100}$. Hence the proposed scheme is secure against this kind of brute-force attacks. We remark that, one can achieve any higher key space by adjusting the parameter values.

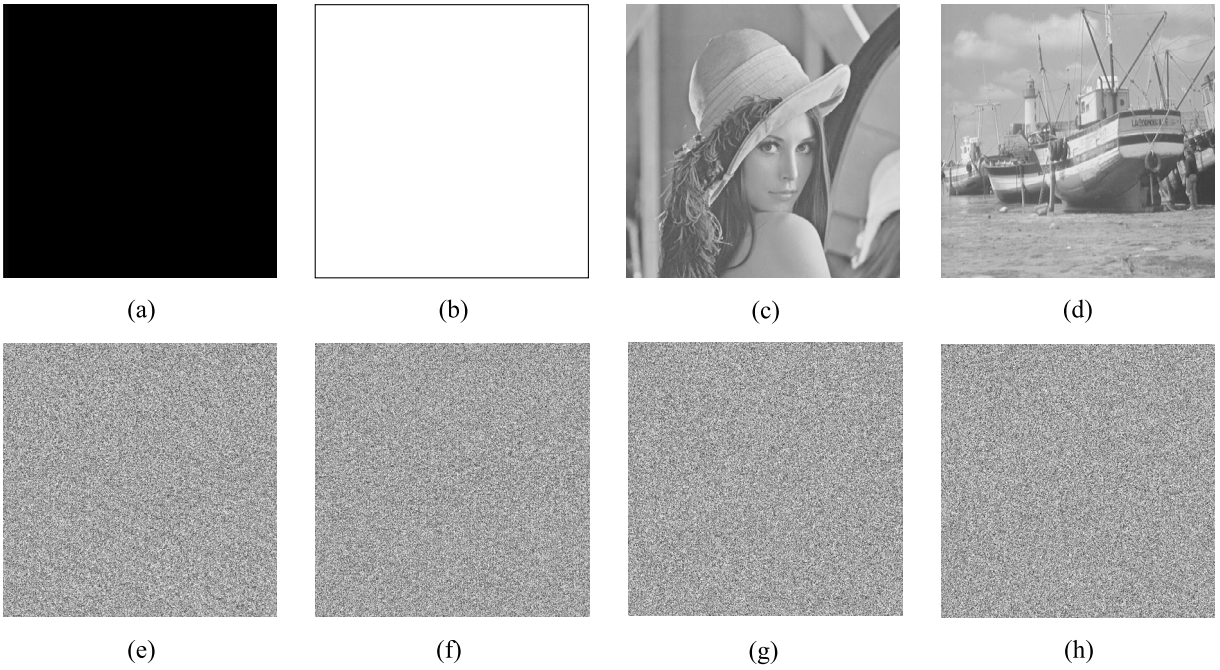



**Fig. 4.** Decrypting cipher-images with a slightly modified secret key: (a)–(d) Graysacle plain-images All-black, All-white, Lena, and Boat with dimension $512 \times 512$; (e)–(h) Decrypted images of the cipher-images of the plain-images in (a)–(d), respectively, with a modified secret key $(x^*, z)$.

### 5.5.2. *Key sensitivity*

For a cryptosystem, it is necessary to create a noticeably different cipher-image when the secret key is slightly changed. To show the sensitivity of our scheme, we decrypted all cipher-images obtained by the proposed scheme with a slightly changed secret key $(x^*, z)$ that is obtained by randomly changing one bit in $y^*$, where $(x^*, y^*)$ is the original secret key. The change in $z$ from $y^*$ is depicted in a box below:

$z =$ F4BB8CC7F86DB26700A7F3EC F EEED3F0B5C6B5107C4DA97740AB21A29906C42DBBB3E377DE9F251F6B93937FA99A3248F4EAFCB
E95EDC0F4F71BE356D661F41B02

The decryption results for the plain-image All-black, All-white, Lena, and Boat, are given in Fig. 4. The result implies that decrypting a cipher-image with a slightly perturbed secret key results in an image that is completely different from the original plain-image. Furthermore, to quantify the effect of a change in the secret key, we computed NPCR and UACI for the cipher-images of the grayscale images and color images obtained from the secret keys $(x^*, y^*)$ and $(x^*, z)$. These results are listed in Tables 4 and 5. For the cipher-images of grayscale and color images, the range (resp., average) of the NPCR is [99.5834, 99.6315] (resp., 99.6097) and [99.5514, 99.6765] (resp., 99.6128), respectively, which is in the recommended values of NPCR. Similarly, the range (resp., average) of UACI for the cipher-images of grayscale and color images, is [33.3068, 33.6499] (resp., 33.4648) and [33.3013, 33.6569] (resp., 33.4740), respectively, which is in the recommended range of values of UACI. This implies that the proposed scheme is highly sensitive to the secret key.

## 5.6. *Analysis of computational complexity*

Computational complexity is one of the critical measures to decide if a cryptosystem can be used in a practical scenario or not. Therefore it is necessary for a cryptosystem to generate cipher-images with low computational complexity. We analyzed the average running time in seconds over all grayscale (resp., color) images listed in Table 4 (resp., Table 5) of each phase of our scheme with the above listed parameters in Table 9, where $\tau_i$, $i = 1, 2, 4, 5$ denotes the average running time to execute phase (I), (II), (IV), (V), respectively; $\tau_{3,1}$ and $\tau_{3,2}$ denote the average running time taken by the solver to generate S-boxes and create confusion in phase (III), respectively; $\tau$ denotes the average total execution time of all phases where for a color image we add the computation time for the generation of key random numbers and S-boxes only once since we use the same key for the three layers; $\tilde{\tau}$ denotes the average execution time excluding the running time for the generation of a key, random numbers and S-boxes. We remark that analyzing the running time $\tilde{\tau}$ is meaningful since large keys are kept fixed in some of the existing cryptosystems, see for example [4,13].

From Table 9 it is evident that all phases have total execution time of less than one second. Similarly the total encryption time $\tau$ is at most 1.5887 seconds, and the total encryption time $\tilde{\tau}$ with a fixed key is at most 0.6256 sec., which is very low. In most of the encryption schemes, the time to encrypt a color image is around $3T$, where $T$ is the time to encrypt one channel/grayscale image.

**Table 9**
Analysis of encryption time in seconds for grayscale and color images.

| Image | Square dimension | $\tau_1$ | $\tau_2$ | $\tau_{3,1}$ | $\tau_{3,2}$ | $\tau_4$ | $\tau_5$ | $\tau$ | $\tilde{\tau}$ |
|---|---|---|---|---|---|---|---|---|---|
| Grayscale | 256 | 0.1028 | 0.3448 | 0.5266 | 0.0043 | 0.0063 | 0.0046 | 0.9893 | 0.0151 |
| | 512 | 0.1011 | 0.3369 | 0.5225 | 0.0154 | 0.0181 | 0.0189 | 1.0128 | 0.0524 |
| | 1024 | 0.1028 | 0.3373 | 0.5159 | 0.0583 | 0.0703 | 0.0755 | 1.1600 | 0.2041 |
| Color | 256 | 0.0917 | 0.3470 | 0.5183 | 0.0116 | 0.0179 | 0.0133 | 0.9997 | 0.0428 |
| | 512 | 0.0948 | 0.3514 | 0.5220 | 0.0481 | 0.0569 | 0.0572 | 1.1303 | 0.1622 |
| | 1024 | 0.1040 | 0.3434 | 0.5157 | 0.1816 | 0.2142 | 0.2298 | 1.5887 | 0.6256 |

**Table 10**
Comparison of running time of the proposed scheme with the schemes [4,13].

| Method | Square dimension | $\lvert\text{bin}(p)\rvert$ | Time (sec.) | Avg. rate of change |
|---|---|---|---|---|
| Ref. [4] | 256 | 21 | 0.1526 | **2.08E-07** |
| | 512 | 21 | 0.1946 | |
| | 1024 | 21 | 0.3571 | |
| Ref. [13] | 256 | 17 | 5.0637 | **1.54E-03** |
| | 512 | 19 | 79.1797 | |
| | 1024 | 17 | 1516.9938 | |
| Proposed ($\tau$) | 256 | 521 | 0.9893 | **1.74E-07** |
| | 512 | 521 | 1.0128 | |
| | 1024 | 521 | 1.1600 | |
| Proposed ($\tilde{\tau}$) | 256 | 521 | 0.0151 | **1.92E-07** |
| | 512 | 521 | 0.0524 | |
| | 1024 | 521 | 0.2041 | |

However, from Table 9 we observe that the total encryption time $\tau$ for color images is almost the same as that for the grayscale images which is much less than $3T$. Hence, the proposed scheme is a good candidate for real-time encryption of grayscale and color images. To show the computational efficiency of the proposed scheme, we compared the encryption speed of the proposed scheme with EC-based encryption schemes [4,13] under the same environment as explained in Section 5, and for the same data set listed in Table 4. For the scheme [13], we used the prime number $p = 66973$, 264529, and 1050737 as a secret key to encrypt images of dimension $256 \times 256$, $512 \times 512$, and $1024 \times 1024$, respectively. For the scheme [4], we used the prime $p = 1048847$ for encryption. The results of these experiments are listed in Table 10, where we also list the average rate of change in running time (the difference of running time taken by the image of size $1024 \times 1024$ and the time taken by the image of size $256 \times 256$ divided by the difference of their numbers of pixels) to understand the effect of the change in the size of plain-image on the encryption time. It is evident that the proposed scheme is much faster than the scheme [13] even when the underlying prime is much larger than that of the scheme [13]. From the computational results, it can be observed that the scheme [4] is slightly faster than the proposed scheme when the underlying keys are not fixed. However, there is a huge difference between the underlying primes and we expect that the proposed scheme will efficiently perform the encryption when compared to the scheme of Azam et al. [4] with the same underlying primes. But it is almost impossible to use the scheme [4] with such a large prime without some modifications since this scheme requires the storage of the underlying EC. Furthermore, our scheme is computationally more efficient than the scheme [4] when the underlying keys are fixed. The average rate of change in running time of the proposed scheme is smaller than the schemes [4,13]. This indicates that the proposed scheme is efficient when compared with the schemes [4,13].

Generally, the computation complexity of an encryption scheme increases by a constant factor with the size of the plain-image. More precisely, if $T$ is the running time for an encryption scheme to encrypt a plain-image of size $i$, then the time to encrypt a plain-image of size $\lambda i$ by the same scheme should be upper bounded by $\lambda T$, where $\lambda$ is a positive scalar. Recall that we used images of size $256^2$, $512^2 = 4 \cdot 256^2$ and $1024^2 = 4 \cdot 512^2$ in our experiments. Thus the encryption time for an image of size $512^2$ and $1024^2$ should be bounded above by $4\tau_{256}$ and $4\tau_{512}$, respectively, where $\tau_i$ is the encryption time for images of size $i^2, i = 256, 512$. The computational efficiency of a scheme is inversely proportional to $\tau_{2i}/\tau_i$, $i = 256, 512$, and therefore we examine the value of $\tau_{2i}/\tau_i$ as our measure of efficiency. For the schemes [4,6,10,11,13,14,16,17,20,23,25,27,43] and the proposed scheme we computed $\tau_{2i}/\tau_i$, $i = 256, 512$ in Table 11. The running time $\tau_i, i = 256, 512, 1024$ for the listed schemes is taken from their respective references, and therefore the underlying experimental environment might be different from the proposed scheme. It is worth mentioning that it is logical to compare $\tau_{2i}/\tau_i$ for different schemes even when the underlying environments are different, since for a fixed scheme $\tau_i$ is computed over the same environment. From Table 11, we observe that the proposed scheme has the minimum value of $\tau_{2i}/\tau_i$, $i = 256, 512$ among the corresponding values for all the listed schemes. This further justifies that the proposed scheme is more suitable for image encryption.

**Table 11**
Comparison of the difference in running time of the proposed scheme with some of the existing schemes.

| Image | Method | Time (sec.) | | | | |
|---|---|---|---|---|---|---|
| | | $\tau_{256}$ | $\tau_{512}$ | $\tau_{512}/\tau_{256}$ | $\tau_{1024}$ | $\tau_{1024}/\tau_{512}$ |
| Grayscale | Ref. [4] | 0.1526 | 0.1946 | **1.2757** | 0.3571 | **1.8346** |
| | Ref. [6] | 0.095 | 0.497 | **5.2316** | 2.513 | **5.0563** |
| | Ref. [11] | N.A. | 0.969 | N.A. | 3.76 | **3.8803** |
| | Ref. [13] | 5.0637 | 79.1797 | **15.6366** | 1516.9938 | **19.1589** |
| | Ref. [14] | 0.0334 | 0.1812 | **5.4251** | 0.8253 | **4.5546** |
| | Ref. [16] | 0.071 | 0.313 | **4.4085** | 1.947 | **6.2204** |
| | Ref. [20] | 0.0168 | 0.0193 | **1.5337** | 0.0296 | **3.9524** |
| | Ref. [23] | 0.0518 | 0.1249 | **2.4112** | N.A. | N.A. |
| | Ref. [25] | 0.0724 | 0.3569 | **4.9296** | 1.6148 | **4.9296** |
| | Ref. [27] | 0.0096 | 0.0239 | **2.4886** | 0.0693 | **2.8996** |
| | Proposed ($\tau$) | 0.9893 | 1.0128 | 1.0238 | 1.1600 | 1.1453 |
| Image | Method | Time (sec.) | | | | |
| | | $\tau_{256}$ | $\tau_{512}$ | $\tau_{512}/\tau_{256}$ | $\tau_{1024}$ | $\tau_{1024}/\tau_{512}$ |
| Color | Ref. [10] | 2.2636 | 8.4003 | **3.7100** | 34.3617 | **4.0905** |
| | Ref. [14] | 0.0800 | 0.4842 | **6.0525** | 2.2848 | **4.7187** |
| | Ref. [16] | 0.2483 | 1.1386 | **4.5856** | 4.8970 | **4.5546** |
| | Ref. [17] | 0.1253 | 0.5101 | **4.9296** | 2.3802 | **4.6661** |
| | Ref. [43] | 2.2786 | 8.4127 | **3.6920** | 34.5904 | **4.1117** |
| | Proposed ($\tau$) | 0.9997 | 1.1303 | 1.1306 | 1.5887 | 1.4055 |

### 5.7. Discussion

We discuss the key features of our scheme as follows.

(1) The proposed scheme generates dynamic S-boxes that have deterministic nonlinearity, which is not the case in the existing encryption schemes including the schemes in [3,4,7,12,13,18–20,28,32,33]. Intuitively, this fact guarantees a high level of security, which is not the case in the existing schemes.
(2) The proposed scheme can create highly secure cipher-images by generating and then using a small number of random numbers, which is not possible in some of the existing schemes. For example, the schemes [4] and [13] are required to generate a sequence of as many random numbers as the size of the plain-image, which eventually has bigger computation cost compared to our new scheme.
(3) For an image of size $u \times v$, the scheme [13] is required to generate an ordered EC and the scheme [4] needs to store an ordered EC of size at least $uv$. This is time-consuming and can create a storage problem as discussed in Section 5.6. However, the proposed scheme does not have such conditions of ordering and on the size of the underlying EC and can work with ECs with small size. This fact makes our scheme more flexible and efficient, as shown with the experimental results in Table 10.
(4) Some EC-based schemes including [4] and [13] cannot directly work with large primes since they generate ECs, and therefore their key space is small. On the other hand, our proposed scheme can efficiently work with arbitrarily large primes, which makes the proposed scheme more reliable while generating a secret key.
(5) The proposed scheme has better resistance against differential attacks compared to the schemes [13–15,24,25,30,35,43] (see Table 3).
(6) The proposed scheme can generate cipher-images with high confusion and diffusion as compared to the schemes [14,20,22–25, 29,30,35,37] (see Table 8).
(7) The proposed scheme can efficiently encrypt both grayscale and color images, with no significant difference in running time (see Table 9). Compared with several existing schemes, including [4] and [13], our new scheme is computationally highly efficient when (see Table 10). Furthermore, in the proposed scheme, the encryption time scales favorably with the size of the input image, evident when compared with the schemes [4,6,10,11,13,14,16,17,20,23,25,27,43] (see Table 11). Hence the proposed scheme is a good candidate for real-time image encryption.

## 6. Conclusion

We proposed a novel public-key substitution-diffusion image encryption scheme with an S-box generator which has provable security against linear attacks. This scheme has five main phases (I)–(V): phase (I) is to generate public and secret keys where we used the Diffie-Hellman protocol to generate secret keys; phase (II) is to create random numbers by using isomorphic ECs; phase (III) is to construct dynamic S-boxes with a given nonlinearity based on ILPs, and then these S-boxes are used to create confusion in a given plain-image; phase (IV) is to create diffusion in a given image by propagating a change in the pixels using random numbers and S-boxes; and phase (V) is to generate a cipher-image by using a mapping between points on isomorphic ECs and pixel values of an image. We performed rigorous analysis on our S-box generator and encryption scheme. We gave an ILP formulation together with a proof that it is feasible if and only if there exists an S-box with a given nonlinearity, and moreover,

such an S-box can be constructed from a feasible solution to the ILP formulation. Due to this fact our S-box generator guarantees the nonlinearity of the resulting S-boxes which is not the case for almost any of the existing S-box generators including the generators introduced in [3,4,7,12,13,18–20,28,32,33]. Computational experiments confirm that the S-box generator can efficiently generate highly dynamic S-boxes with relatively small size. Furthermore, we have compared the proposed encryption scheme with the existing schemes, from which it is evident that our scheme (i) is more flexible, has large key size, and uses small sequences of random numbers compared with the schemes introduced in [4] and [13]; (ii) has better resistance against differential attacks compared with the schemes [13–15,24,25,30,35,43]; (iii) can generate cipher-images with a high level of confusion and diffusion compared with the schemes [14,20,22–25,29,30,35,37]; and (iv) has better computational efficiency than the schemes available in the literature [4, 6,10,11,13,14,16,17,20,23,25,27,43]. Thus, the proposed scheme can create cipher-images with low computational complexity and a high level of resistance against modern attacks.

We focused on generating S-boxes with a given nonlinearity as a first step towards S-box generators with provable security against linear attacks. An extension of this work would be developing an S-box generator that has provable security against linear and differential attacks that can be used in encryption schemes.

## CRediT authorship contribution statement

Conceptualization, N.A.A. and A.S.; methodology, N.A.A.; validation, N.A.A., J.Z. and U.H.; software, J.Z., U.H., and N.A.A., writing, original draft, N.A.A., and U.H.; writing, review and editing, N.A.A., U.H., and A.S, project leader, N.A.A.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary material

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